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CropCode: Inverse Procedural Modeling of Occluded Row Crop Geometry from Sparse Multimodal Field Observations

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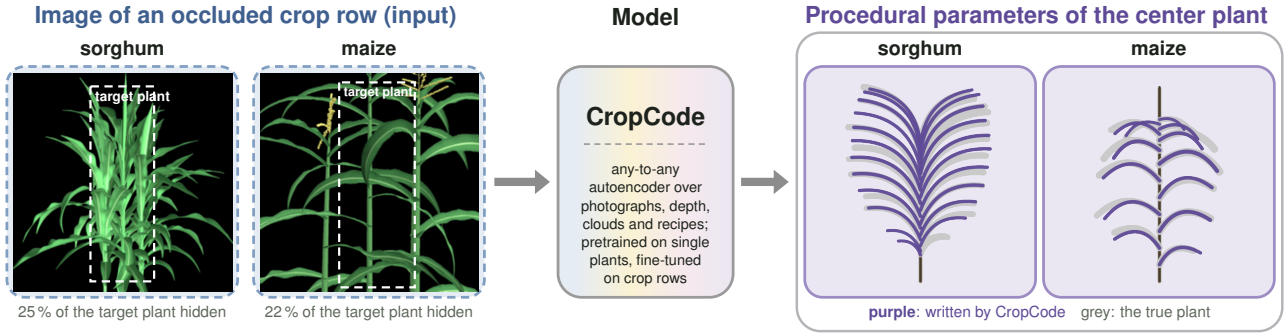


Figure 1. **Any sensor in, procedural plant out.** Given any subset of field-practical inputs—a single RGB image, a depth frame, or a point cloud heavily occluded by neighboring plants—CROPCODE predicts the complete procedural representation of the target crop plant. This compact, kilobyte-scale parameter set encodes biologically meaningful traits and can be re-rendered into full 3D geometry for simulation and field-scale phenotyping. Here we show one held-out simulated test row per species viewed broadside from a rover camera (the dashed box highlights the target plant for visualization; the model receives no bounding box or mask). Taking only the row photograph as input (a depth frame or point cloud works equally well), the row-trained model reconstructs the target plant's procedural parameters, illustrated as a stem schematic showing each leaf's insertion height, length, width, and orientation (purple: predicted parameters; grey: ground truth). Predicted plant height falls within 2% of the true value for both species, and leaf count is accurate within one leaf (Regime III, Sec. 4.4).

Abstract

Editable, biologically interpretable 3D models of field-grown crops are essential for high-throughput phenotyping, in silico *breeding trials, and agricultural digital twins. However, recovering these models from field sensors remains difficult because plant leaves are thin and slender, neighboring plants in a row heavily occlude one another, and routine field campaigns typically capture only a single photograph, a noisy depth map, or a partial point cloud per plant. In this regime of slender geometry under heavy occlusion, both standard 3D point-cloud completion and Chamfer-based evaluation break down. To overcome these challenges, we introduce* CROPCODE*, a multimodal masked-autoencoder framework that recovers the complete procedural representation of a plant—the generator parameters specifying leaf count, insertion heights, blade angles, and surface geometry—from any subset of RGB, depth, and point-cloud observations in a single forward pass.*

We pretrain CROPCODE *on 15,000 procedurally generated plants per species (maize and sorghum), fine-tune it on simulated crop rows where neighboring leaves bend realistically upon contact, and distill a full-modality teacher into single-sensor student models. Cross-modal distillation cuts sorghum parameter error from point clouds alone in half and reduces cross-modal generation loss by 30% in sorghum and 44% in maize. Fine-tuning on simulated rows removes 92% (maize) and 85% (sorghum) of the reconstruction error introduced by occluding neighbors. Through controlled ablations across both species, we establish a key architectural principle: exposing ground-truth procedural parameters to the encoder degrades learned sensor representations, whereas using them strictly as a reconstruction target helps maize while hurting sorghum. We also show that Chamfer distance and linear probes on frozen representations rank sensor modalities differently on thin foliage, motivating our adoption of frozen-latent probes as the primary metric for architectural decisions. Finally, on crop rows constructed from real plants of both species, zero-shot transfer degrades on plants taller than the synthetic training range, but fine-*

tuning on 75% of the real plants brings the held-out test set to a 1.6% median height error and predicts leaf count within one leaf on 70% of maize captures—running in milliseconds per plant rather than minutes or hours. We will release FIELDSCAN*, our 30,000-plant, two-species benchmark pairing multimodal field captures under simulated row occlusion with exact procedural ground truth.*

1. Introduction

Accurate 3D models of individual crop plants allow breeders and agronomists to measure morphological traits—such as leaf angle, internode length, and canopy structure—at scale, while powering agricultural digital twins for *in silico* breeding trials [19, 25]. Procedural representations are uniquely well suited to these applications: instead of storing millions of unstructured points, a procedural model represents a plant as a compact set of biologically meaningful parameters that a generator expands into complete, editable, simulation-ready 3D geometry [20, 36]. A procedural plant requires only a few kilobytes of storage instead of megabytes, maps directly to agronomic traits, and can be cleanly edited or perturbed in ways that raw meshes and point clouds cannot. From a computer vision perspective, recovering such models from field sensors also poses a fundamental 3D reconstruction problem: inferring a parametric model of a thin, heavily occluded object from sparse, partial multimodal observations.

Despite their utility, existing methods for recovering procedural plant models rely on clean, high-density observations of isolated plants. Manual specification takes hours per plant [20]; staged pipelines combining vision-language models with differentiable surface fitting reach 5.4 mm median accuracy, but require a complete terrestrial laser scan of an isolated plant and roughly ten minutes of processing per instance [18]; per-scene optimization fits parameters against multi-view imagery [54]; and existing feed-forward networks assume clean, unoccluded point clouds [8, 14]. High-throughput field phenotyping presents the exact opposite setting: thousands of plants are imaged daily using whichever single sensor or partial sensor suite is practical on a ground rover. Moreover, crop plants consist of slender organs (a maize leaf is only centimeters wide and millimeters thick, yet up to a meter long) tightly interleaved with neighboring plants in the row. In our field simulations, roughly one third of the target plant's visible pixels and two thirds of the captured 3D points belong to adjacent neighbors. This combination violates the core assumptions of standard 3D completion benchmarks, which focus on compact, mostly convex objects [51, 53], and exposes the insensitivity of Chamfer distance to thin sheets [43, 49]. While recent occlusion-aware plant completion methods recover missing geometry, they output unstructured point clouds rather than editable procedural models [27]. To our knowledge, no prior approach recovers a complete, re-renderable procedural plant from a single field observation under row-neighbor occlusion.

CROPCODE makes this inverse problem tractable by using the procedural generator itself as both a data engine and a structural prior. As illustrated in Fig. 2, we pretrain a four-stream masked autoencoder [3, 10, 21, 31] over RGB images, depth maps, point clouds, and tokenized procedural parameters across 15,000 generated plants per species (maize and sorghum). Because random subsets of tokens are masked across modalities during pretraining, a single network learns to infer missing streams from any available sensor combination. We then distill a full-modality teacher [22] into single-sensor students and fine-tune the model on simulated crop rows whose neighboring leaves bend naturally on contact. Cross-modal distillation cuts sorghum parameter error from a point cloud alone in half and reduces overall cross-modal generation loss by 30% in sorghum and 44% in maize, while row fine-tuning eliminates 92% (maize) and 85% (sorghum) of the reconstruction error caused by neighboring plants.

Our controlled ablations also reveal an important architectural insight for any multimodal system that has access to ground-truth generative parameters during training: allowing the encoder to see most of the parameter tokens degrades the learned sensor representation in both species. When parameters are restricted strictly to a decoder reconstruction target under matched token budgets, their effect depends on the underlying generator—degrading sorghum representations while improving maize. Even so, supervising parameters alongside visual reconstruction allows our distilled generalist to outperform a parameter-only specialist at parameter regression (achieving a recipe MAE of 0.019 vs. 0.036 from point clouds alone), consistent with findings on modality competition [35] and negative gradient transfer [50]. Furthermore, we show that while adding depth reduces Chamfer distance by 25–28%, linear probes on frozen representations reveal minimal gain from depth and identify RGB as the primary carrier of morphological trait information. Because Chamfer distance is insensitive to missing thin blades, we adopt frozen-latent probes as our deciding metric [6, 49].

We validate our approach on crop rows assembled from real maize and sorghum plants with independently reconstructed ground truth [18, 25]. While zero-shot accuracy drops on real plants that exceed the synthetic generator's height range, fine-tuning on 75% of the real plants brings the held-out test set to a 1.6% median height error and predicts leaf count within one leaf on 70% of maize captures, running in milliseconds per plant. Our main contributions are:

1. **Task and framework.** We introduce the first amortized framework that recovers complete, editable procedural plant models of slender, row-occluded crops from *any subset* of RGB, depth, and point-cloud inputs in a single forward pass. Full-to-single-sensor distillation reduces generation loss by 30% in sorghum and 44% in maize

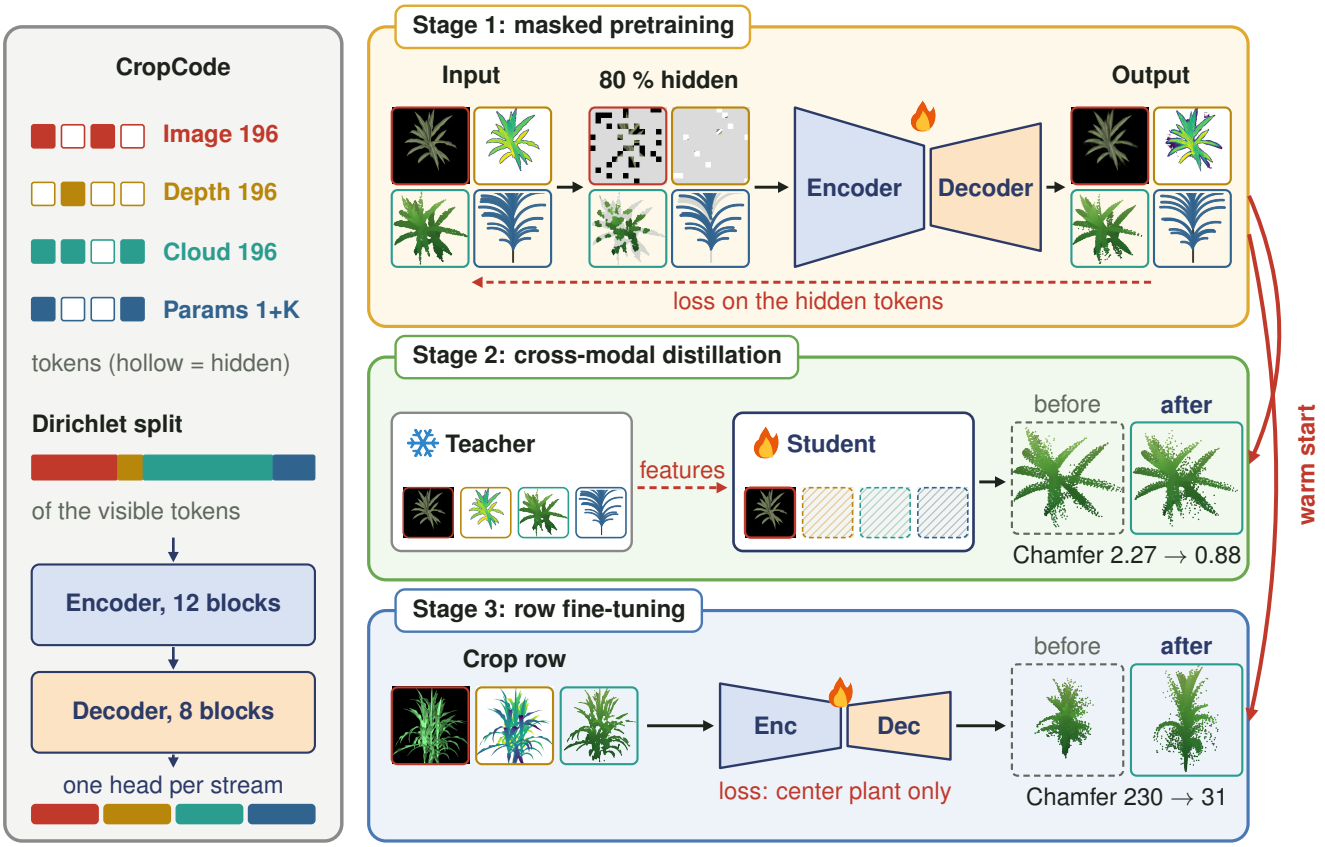


Figure 2. **Overview of the three-stage training pipeline in CROPCODE. Stage 1 (Multimodal Masked Pretraining):** All four plant streams (RGB, depth, point cloud, and procedural parameters, depicted as a stem-and-leaf schematic) are tokenized. We mask 80% of tokens using a random Dirichlet allocation across streams, and a shared encoder–decoder reconstructs all streams (applying image, depth, and parameter losses on masked tokens, and a combined QAL + Sinkhorn loss on the full point cloud). **Stage 2 (Cross-Modal Distillation):** A frozen four-stream teacher transfers its latent representation and decoder features to single-sensor students; we show a point cloud reconstructed from a single RGB image before and after distillation. **Stage 3 (Row Occlusion Fine-Tuning):** Starting from Stage 1 weights, the model is fine-tuned on simulated crop rows with supervision computed strictly against the occluded center plant, enabling it to separate the target plant from its neighbors. Reported Chamfer values are sorghum test means from Table 8 ($\times 10^{-3}$) and Table 7 (all streams, $\times 10^{-4}\,\mathrm{m}^2$) in Appendix A, illustrated on a held-out sorghum test plant and row.

while halving sorghum point-cloud parameter error, and row fine-tuning eliminates 85–92% of neighbor-induced occlusion error.

2. **Design principle for privileged generative modalities.** Through budget-matched ablations across two species, we show that exposing ground-truth procedural parameters to the encoder harms visual representation quality, whereas using them as a decoder-only target helps or hurts depending on the generator's parameter coupling.
3. **Evaluation protocol for slender geometry.** We demonstrate that Chamfer distance and downstream trait probes systematically disagree on thin plant organs, and we establish a frozen-latent probing protocol to guide architectural and modality choices where nearest-neighbor geometric metrics fail.
4. **The FIELDSCAN benchmark.** We will release FIELDSCAN, a 30,000-plant, two-species benchmark pairing multimodal sensor captures and physically simulated row occlusion with exact procedural ground truth, alongside a real-field evaluation split across a maize diversity panel [25] (Table 2).

2. Related Work

We briefly review the four areas most closely related to CROPCODE, with an expanded literature survey provided in Appendix F.

Recovering procedural plant models. Classical inverse procedural modeling estimates tree generator parameters on a per-instance basis using stochastic optimization [44], MCMC sampling [45], or L-system induction [17, 28]. More recent crop-focused methods infer parametric morphology

from clean, complete 3D point clouds [8, 14], optimize procedural parameters against multi-view NeRF reconstructions [54], or combine vision-language prompting with differentiable NURBS fitting on terrestrial laser scans [9, 18]. However, every existing approach assumes a clean, isolated plant scan or requires minutes of per-plant optimization, leaving missing modalities and dense row-neighbor occlusion unaddressed.

Inferring programs from visual observations. Outside of agriculture, feed-forward models have been trained to infer 3D shape programs [46], diffusion-based procedural assets [55], and executable CAD command sequences [24, 26] from images or point clouds. These systems target rigid, isolated man-made objects under clean viewing conditions and do not examine whether continuous program parameters should be fed into the encoder during multimodal pretraining.

Point-cloud completion and slender geometry. Deep point-cloud completion networks [5, 51–53] are primarily developed on compact, mostly convex shapes. Recent work on occluded plant completion [27] and thin-structure reconstruction [13, 39] handles slender geometry, but outputs unstructured point sets rather than editable, trait-level procedural models. Moreover, standard Chamfer distance is structurally biased on thin surfaces because clustering points along thick stems yields low nearest-neighbor error even when entire leaves are missing [30, 43, 49]. We address this by pairing distribution-matching losses with frozen-latent trait probes (Sec. 4.7).

Multimodal masked pretraining and synthetic data. Masked autoencoding [21] has been extended to multimodal images [3], any-to-any generation [4, 31], 3D point clouds [34], and joint RGB–depth–point-cloud inputs in EmbodiedMAE [10], which serves as our architectural backbone. We introduce a fourth structured modality—continuous procedural parameters embedded via xVal-style numerical tokens [15]—and distill a full-modality teacher [22] into single-sensor students to handle missing modalities at test time [41]. While pretraining on procedural synthetic data is well established [11, 16, 23, 40, 47], our setting is uniquely aligned because the generator defines the exact target output space and is validated directly against real-field scans [25].

3. Method

We first formalize inverse procedural plant reconstruction (Sec. 3.1) and describe our FIELDSCAN synthetic data engine (Sec. 3.2). We then detail our four-stream masked pretraining (Sec. 3.3), cross-modal teacher–student distillation (Sec. 3.4), and field evaluation protocols.

Table 1. **FIELDSCAN dataset configuration.** Parameter dimensions combine plant-level macro traits with K structured leaf slots. Each row scene places a target plant between two neighboring plants spaced at 0.15 m stem intervals.

Parameter / Configuration	Maize	Sorghum
Total generated plants	15,000	15,000
Split (train / val / test)	10,500 / 2,250 / 2,250	
Views per plant	10 (random sphere)	10 (fixed rig)
Camera distance	box-fitted	2.5 m
Per-view point cloud	8,192 pts (visible)	complete plant cloud
Parameter dim. d_θ	$14{+}14{\times}28{=}406$	$9{+}5{\times}24{=}129$
Leaf count range	3–22	4–35
Plant height range	0.62–1.44 m	0.19–1.74 m (stem)
Row scenes × orientations	$15{,}000 \times 10$	$15{,}000 \times 10$
Row camera	level rover, 1.0 m	level rover, 2.5 m
Row cloud (input / target)	24,576 / 8,192 pts	
Real-field split	520 real plants, 2 layouts	30 scanned plants, 2 layouts

3.1. Problem Formulation: Inverse Procedural Reconstruction

Let G denote a species-specific procedural plant generator that maps a compact parameter vector $\theta \in \mathbb{R}^{d_\theta}$ to an editable 3D mesh $G(\theta)$. The vector θ provides a complete structural recipe for the plant, specifying global traits such as stem height and internode spacing alongside per-leaf attributes including insertion height, blade length and width, and insertion, roll, and azimuth angles.

In practical field campaigns, rover-mounted sensors produce partial, noisy, and occluded observations from a subset of modalities $S \subseteq \mathcal{M} = \{\mathrm{rgb}, \mathrm{d}, \mathrm{pc}\}$, where rgb denotes a color image, d a metric depth map, and pc a 3D point cloud. Our goal is to learn an amortized feed-forward inverse mapping,

$$f_\phi(\{x^m : m \in S\}) = \left(\hat{\theta}, \{\hat{x}^m : m \in \mathcal{M}\}\right), \qquad (1)$$

that predicts both the complete procedural parameter vector $\hat{\theta}$ and the completed sensor streams $\{\hat{x}^m\}$ in a single forward pass. Unlike unstructured point-cloud completion, which must extrapolate hidden surfaces point by point, predicting $\hat{\theta}$ leverages G as an explicit biological prior: re-rendering $G(\hat{\theta})$ produces watertight, structurally valid, and editable 3D geometry in milliseconds rather than minutes or hours of per-plant optimization.

3.2. A Procedural Field-Data Engine: FIELDSCAN

To train and benchmark CROPCODE under controlled yet realistic field conditions, we build FIELDSCAN, a large-scale dataset pairing exact procedural ground truth with multimodal captures of isolated plants and densely planted crop rows (Table 1 and Fig. 3).

Data generation overview. Using species-specific procedural generators [33, 48], we synthesize 15,000 plants each for maize and sorghum. Maize plants span 0.62–1.44 m in height with 3–22 leaves governed by four coupled latent

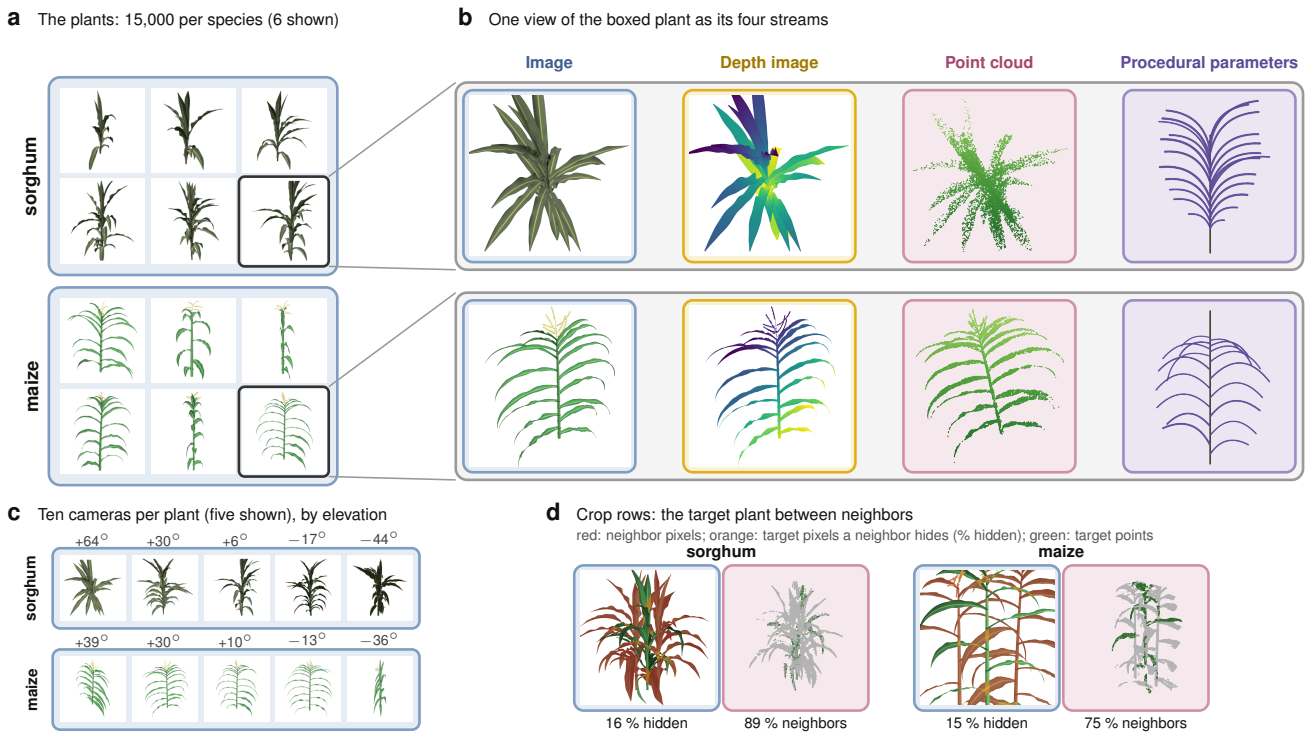


Figure 3. **Overview of the FIELDSCAN benchmark. (a)** Six example validation plants per species shown from a side view, illustrating morphological diversity across the 15,000 generated plants per crop. **(b)** Four aligned data streams for a single camera view of the highlighted plant: RGB image, metric depth map, camera-frame 3D point cloud, and procedural parameters visualized as a stem-and-leaf schematic. **(c)** Five of the ten camera viewpoints per plant annotated by elevation angle: sorghum uses a fixed elevation ladder from $+64°$ to $-64°$, whereas maize cameras are sampled uniformly on the sphere (sorted here by elevation). **(d)** Simulated crop rows at $0°$ yaw showing heavy inter-plant occlusion: in the RGB view, neighbor pixels are highlighted in red and target pixels hidden behind neighbors in orange; in the point cloud, target points are shown in green and occluding neighbor points in grey.

traits, leaving height and leaf count only weakly correlated ($r = 0.20$). In contrast, sorghum plants (generated via EvoEngine) have stem lengths of 0.19–1.74 m and 4–35 leaves tightly coupled to stem length ($R^2 = 0.988$). Together, the two species test CROPCODE across both weakly and strongly coupled morphological regimes.

For isolated-plant pretraining, we render ten 1024×1024 views per plant ($40°$ field of view) with RGB imagery, losslessly packed metric depth, and camera calibration. Sorghum is captured from a fixed 10-camera Fibonacci rig at 2.5 m with complete 8,192-point plant clouds per view, whereas maize is captured from random spherical directions at bounding-box-fitted distances with partial 8,192-point clouds back-projected directly from the depth map. To simulate field rows, each target plant is placed between two neighbors spaced 0.15 m apart across ten row orientations ($0°$–$90°$ yaw, with $\pm 15°$ random neighbor azimuth jitter). An articulated contact solver models leaf blades as chains of 8 cm jointed segments and resolves all triangle–triangle collisions so neighbors yield first while the target plant retains its grown shape wherever possible. A simulated ground-rover camera captures the crowded row (RGB, depth, and a 24,576-point scene cloud) paired with the contact-bent target plant alone (8,192 points) as ground truth. In these rows, neighbors account for 59–68% of all plant pixels in maize and occlude 23% of the target plant's pixels in sorghum. We split plants 70% / 15% / 15% into train, validation, and test sets, enriching the validation and test splits toward extreme morphologies via Mahalanobis ranking. Full generation, rendering, and contact-solver details appear in Appendix J.

Table 2. **Comparison with existing 3D plant and shape completion benchmarks.** "Exact proc. GT" denotes exact generative parameters per plant; (✓) indicates parameters fitted post hoc via optimization. †FIELDSCAN includes real-field evaluation scans alongside synthetic training data.

Dataset	Real scans	Slender	Row occl.	RGB+D+PC	Exact proc. GT	Scale
PCN / ShapeNet-55 [51, 53]	×	×	×	×	×	55,000
Pheno4D [42]	✓	✓	×	×	×	84
Crops3D [56]	✓	✓	×	×	×	2,500
MaizeField3D [25]	✓	✓	×	×	(✓)	1,000
FIELDSCAN (ours)	✓†	✓	✓	✓	✓	**30,000**

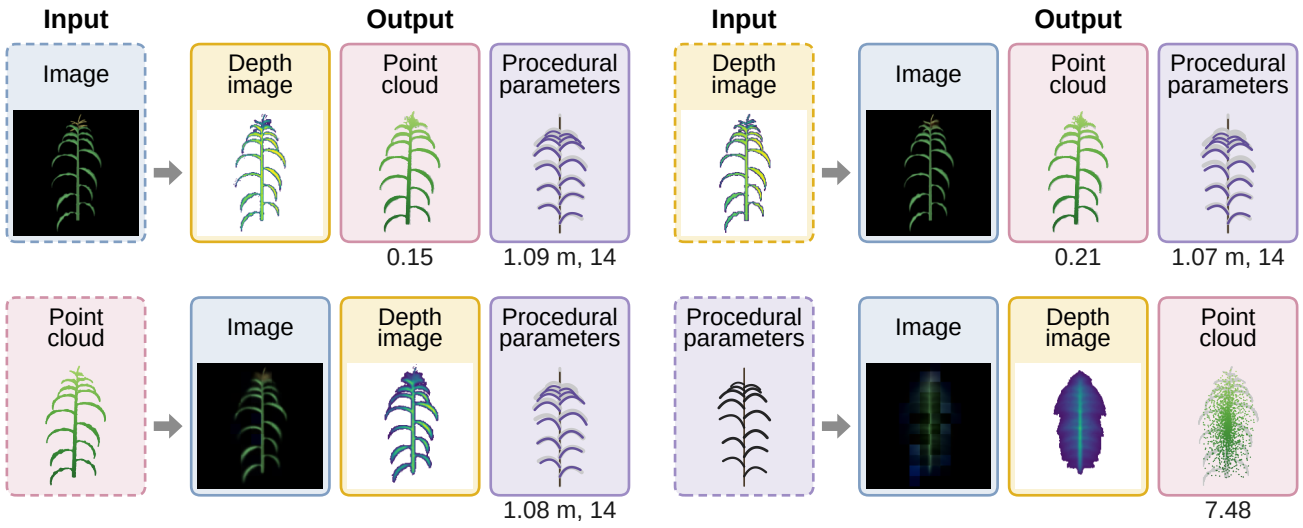


Figure 4. **Any-to-all cross-modal reconstruction on a held-out maize test plant.** In each block, the distilled generalist receives a single input stream (dashed box, left of arrow) and reconstructs the remaining three streams (solid boxes: RGB in blue, depth in yellow, point cloud in pink, and procedural parameters in purple). Predicted point clouds and procedural schematics are overlaid on the ground truth in grey, with Chamfer distance ($\times 10^{-3}$) reported below each cloud and predicted height and leaf count below each parameter schematic (ground truth: 1.12 m, 14 leaves). This example is drawn from test plants with low image-to-cloud error viewed within 25° of the horizon. Reconstructing geometry from parameters alone yields only a coarse approximation (Chamfer 7.48 here; 12.8 on average in Table 8).

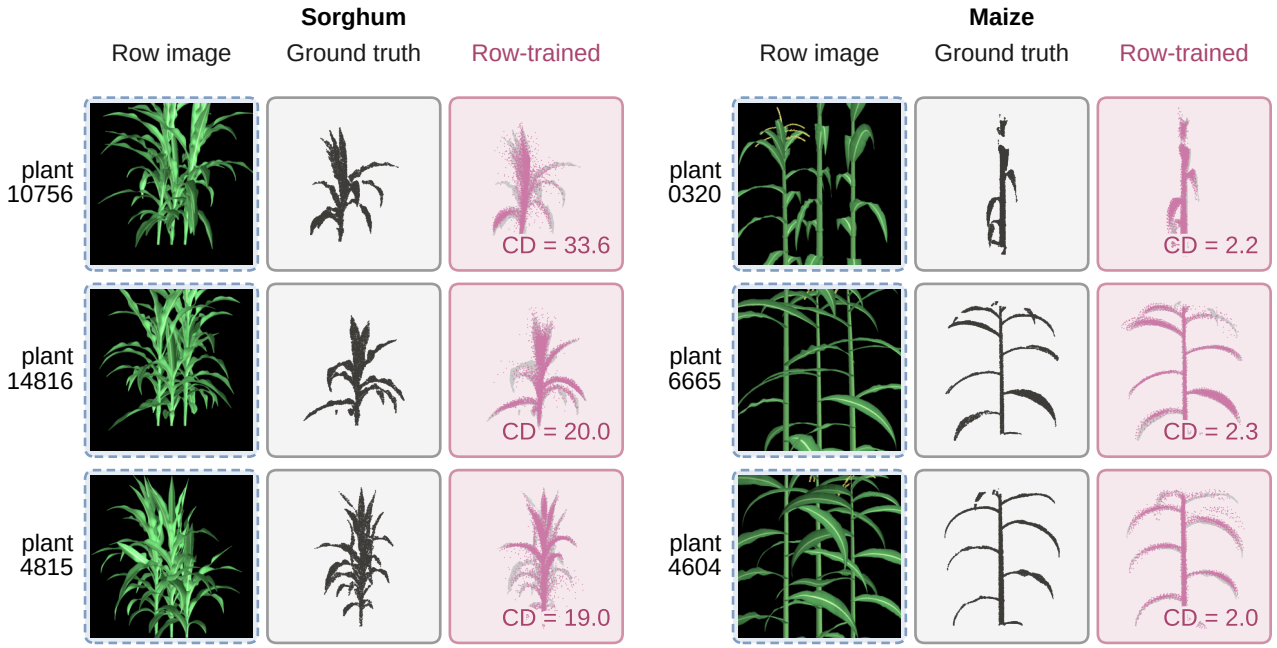


Figure 5. **Reconstructing occluded target plants in simulated crop rows.** Results on held-out sorghum (left) and maize (right) test rows. Columns show the input row photograph (including adjacent neighbors), the ground-truth point cloud of the center target plant, and the point cloud predicted by our row-trained model (pink) overlaid on the ground truth (grey), with metric Chamfer distance ($\times 10^{-4}$ m^2, lower is better) noted below. Examples are selected from plants where the row-trained model ranks in its top 35% and the isolated-trained control ranks in its bottom 40% across both broadside and end-on views; Figs. 28 and 29 show ten additional test plants per species sampled evenly across the full error distribution.

3.3. Four-Stream Masked Pretraining

Multimodal tokenization. Extending EmbodiedMAE [10], we tokenize all four plant modalities into a unified sequence. RGB and depth images are partitioned into $14 \times 14 = 196$ patches and linearly projected to 768-dimensional embeddings with 2D sine–cosine positional encodings. Point clouds are grouped into 196 neighborhoods of 32 points via farthest-point sampling and encoded with a lightweight PointNet [34, 37]. The procedural parameter vector θ is structured into $1 + K$ continuous numerical tokens—one global plant token followed by K ordered leaf slots ($K = 28$ for maize, $K = 24$ for sorghum)—affinely normalized to $[0, 1]$ and embedded using xVal-style continuous value scaling [15]. After adding learned modality embeddings and a global `[CLS]` token, visible tokens are processed by a 12-block ViT encoder (width 768) and reconstructed by an 8-block decoder (width 512) with linear heads for RGB and depth, a FoldingNet head (41 points per token) for point clouds, and an MLP head for procedural parameters. During training, valid leaf slots are selected via a ground-truth existence mask; at test time without ground truth (Table 5, Appendix A), leaf count is inferred directly from the predicted recipe (explicitly stored in the plant token for maize, and derived from predicted stem length for sorghum).

Dynamic Dirichlet masking. To train a single network capable of handling any subset of sensor inputs, we mask a fraction $\rho = 0.80$ of all tokens at each step, leaving $V = \lfloor (1-\rho)L \rfloor$ visible tokens (122 of $L = 613$ tokens in sorghum; 123 of 617 in maize). Visible tokens are allocated across the four streams using a symmetric Dirichlet draw at every iteration, constrained so that each active stream retains at least one visible token and at least 25% masked tokens. In our default configuration, the parameter stream shares this visibility budget ($\sim 58\%$ visible on average), whereas our controlled ablations in Sec. 4.6 gate parameter visibility at a fixed rate ρ_θ (with $\rho_\theta = 1$ making parameters a pure decoder reconstruction target).

Multi-term reconstruction objective. We train the network with a weighted sum of modality-specific reconstruction losses:

$$\mathcal{L}_{\text{rec}} = \ell_{\text{rgb}} + \ell_{\text{d}} + \ell_{\text{pc}} + 5\ell_\theta, \tag{2}$$

where ℓ_{rgb} is the MSE on patch-normalized RGB pixels [21], ℓ_{d} is the MSE on min–max normalized depth, and ℓ_θ is the Smooth-L_1 loss on masked parameter tokens. Because standard Chamfer distance allows decoders to minimize error by clustering points onto thick stems while omitting thin leaf blades [43, 49], we use a sigmoid-weighted quality-aware loss (QAL) that focuses gradients on poorly reconstructed regions:

$$\ell_{\text{pc}} = \frac{1}{|\hat{P}|}\sum_{p\in\hat{P}} w(\delta(p,P))\,\delta(p,P) + \frac{1}{|P|}\sum_{q\in P} w(\delta(q,\hat{P}))\,\delta(q,\hat{P}), \tag{3}$$

where $\delta(p, P) = \min_{q\in P} \|p-q\|_2$ and $w(r) = \sigma(a(r-\tau))$ with sharpness $a = 100$ and tolerance $\tau = 0.01$ in unit-sphere coordinates. For our headline models, we further add a debiased Sinkhorn optimal transport divergence [12] over 1,024-point subsets ($\varepsilon = 0.02$) to encourage global coverage of thin foliage:

$$\ell^\star_{\text{pc}} = \ell_{\text{pc}} + 0.5\, S_\varepsilon\left(\hat{P}_{1024}, P_{1024}\right). \tag{4}$$

All models are trained with AdamW and cosine learning-rate decay. Our headline models train for 1,000 epochs over all ten views (global batch size 256, peak learning rate 6×10^{-4}), whereas controlled ablations share a fixed compute budget of 197,400 steps (global batch size 32, one random view per plant per epoch, peak learning rate 1.5×10^{-4}) using Eq. (3).

3.4. Full-Modality Teacher to Single-Sensor Students

At deployment, a single-sensor model sees only one modality while every token of the remaining streams is absent—a configuration rarely encountered under random Dirichlet masking. To bridge this gap, we distill a frozen four-stream teacher into single-sensor student models. The teacher receives privileged access to the procedural parameters (all parameter tokens visible for sorghum; 50% of each stream visible for maize, where full visibility degrades cloud reconstruction) and produces a global latent vector $c^T \in \mathbb{R}^{768}$ and normalized decoder features $F^T \in \mathbb{R}^{L\times512}$.

Each student is initialized from the teacher's weights and trained for 100 epochs (AdamW, learning rate 10^{-4}, global batch size 128). On 80% of training steps, the student receives only a single active source stream $s \in \{\text{rgb}, \text{d}, \text{pc}, \theta\}$ (with all other streams completely masked); on the remaining 20%, it follows standard four-stream masked pretraining. The student minimizes:

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{rec}} + \frac{\lambda_F}{|\mathcal{J}_{\neg s}|}\sum_{j\in\mathcal{J}_{\neg s}} \|F^S_j - F^T_j\|_2^2 + \lambda_c\|c^S - c^T\|_2^2, \tag{5}$$

where $\mathcal{J}_{\neg s}$ indexes the decoder positions of the unobserved streams, $\lambda_F = 1.0$, and $\lambda_c = 0.5$. At inference, each student operates solely on its designated input sensor.

Field evaluation protocols. We evaluate CROPCODE across three complementary protocols:

1. **Trait probing:** We fit ridge regression probes on frozen `[CLS]` representations using the training split and evaluate R^2 on held-out plants (using one deterministic view per plant with parameter inputs zeroed out to prevent label leakage). A difference is counted as significant only when its 95% paired-bootstrap confidence interval (2,000 plant resamples) excludes zero on both validation and test splits. Under partial point-cloud visibility, probes are

either refit per visibility level (*matched*) or transferred directly from complete clouds (*transferred*).

2. **Row occlusion:** We test both zero-shot transfer of isolated-trained models and row fine-tuned models on simulated crop rows, evaluating reconstruction accuracy strictly against the occluded target plant and verifying via an echo test that models do not merely copy neighboring foliage.
3. **Real-field validation:** We evaluate row models on crop rows assembled from real-world MaizeField3D captures [25], benchmarking against procedural fits produced by a staged vision-language pipeline [18].

4. Experiments

Unless numbered five or lower, the figures referenced in this section appear in Appendix B, and full result tables appear in Appendix A.

4.1. Experimental Setup

Models and compute budgets. All controlled ablations use a ViT-Base backbone (114 M parameters) trained for an identical budget of 197,400 optimizer steps at global batch size 32 (equivalent to 600 epochs of one random view per plant) with matched random view-sampling seeds; dataset-scaling experiments adjust epoch counts accordingly to hold total gradient steps constant. Our headline models train for 1,000 epochs across all ten views at global batch size 256 with the Sinkhorn objective (Eq. 4) and initialize all teacher and student networks in Sec. 4.5. Unless noted otherwise, ablation numbers are reported on the validation split.

Evaluation metrics. Geometric accuracy is measured via symmetric squared Chamfer distance in the normalized unit-sphere frame ($\times 10^{-3}$), alongside point-cloud F_1 score and metric root-mean-square (RMS) error in millimeters. Parameter recovery is reported as normalized mean absolute error (MAE) or trait *skill* ($1 - \mathrm{MAE}/\mathrm{MAE}_{\mathrm{mean}}$, where 1 indicates perfect prediction and 0 matches predicting the dataset mean). Representation quality is evaluated via frozen-latent probe R^2 under the paired-bootstrap protocol above. Full metric definitions and species-specific trait sets are detailed in Appendix C.

4.2. Regime I: Single-View Reconstruction of Isolated Plants

Table 5 (Appendix A) evaluates single-sensor inference on isolated validation plants, where the model receives a single RGB image, depth map, or point cloud and outputs both the procedural parameters and the remaining sensor streams in one forward pass. Across all single-sensor inputs, CROP-CODE recovers the procedural plant reliably: predicted leaf count falls within one leaf of ground truth on 91–100% of plants, height error is 2–5%, and mean leaf-parameter skill reaches 0.66–0.68 in sorghum and 0.57–0.59 in maize. Skill is highest for leaf insertion height and angle, moderate for blade length, and lowest for sorghum leaf width, which exhibits narrow variation across plants. Averaging predicted recipes across three views improves accuracy across every metric. Conversely, a two-stage baseline that first predicts a point cloud from RGB and then infers parameters from the generated cloud loses almost all parameter accuracy (sorghum leaf skill drops from 0.66 to -0.01), confirming the benefit of direct end-to-end parameter prediction.

The reconstructed 3D geometry mirrors these trends. Depth is the strongest single input for point-cloud generation (Chamfer 0.93×10^{-3} in sorghum and 0.64×10^{-3} in maize, corresponding to median RMS errors of 30 mm and 12 mm, and F_1 scores of 0.44 and 0.53), with RGB within a factor of 1.5 and achieving a test-split skill of 0.98–0.99 relative to random plant pairs (Sec. 4.5). Notably, giving the undistilled headline model all three streams performs no better than giving the distilled generalist a single sensor, since the undistilled model was trained with 80% masking rather than full-stream visibility. Qualitatively, Fig. 6 (Appendix B) shows that the distilled generalist resolves crisp, individual leaf blades from a single photograph where the undistilled model produces diffuse point clouds.

4.3. Regime II: Viewpoint Elevation, Self-Occlusion, and Complexity

Effect of camera elevation. Because all ten views of a plant share the exact same ground-truth procedural parameters, evaluating across the elevation ladder isolates the effect of camera angle. Single-image reconstruction error follows a symmetric U-shaped curve across elevation, lowest from horizontal side views and highest when looking steeply down or up (Fig. 7A). Camera elevation accounts for $\eta^2 = 0.24$–0.31 of per-plant variance, whereas morphological traits correlate negligibly ($|\rho| \leq 0.06$). Distillation reduces error by 28% across all viewpoints while preserving the underlying U-shaped difficulty profile.

Training view distribution. Models pretrained on narrow elevation bands (top-down views $i \in \{0, 1\}$, horizontal views $i \in \{4, 5\}$, or a single top-down view $i = 0$) excel only at their trained angles and degrade sharply elsewhere: sorghum per-view probe R^2 drops by 0.066–0.144 on unseen elevations, whereas training across all ten views maintains $R^2 = 0.734$–0.778 everywhere (Fig. 8). In maize, every restricted-view model loses on 10–11 of 11 traits under paired bootstraps, with point-cloud F_1 showing the same pattern. Covering the full range of evaluation elevations thus matters far more than image count alone (Appendix G.1).

Self-occlusion under partial coverage. When point clouds are restricted via mesh ray-casting to surfaces visible from 1 or 3 cameras (49% and 78% surface coverage), latent trait information remains intact if the linear probe is refit to the partial distribution (achieving 0.960 height R^2 from a single

Table 3. **Comparison against published self-supervised encoders (E8).** All encoders are evaluated on identical held-out validation plants using mean-pooled token features (excluding `[CLS]` for CROPCODE) and the same cross-validated ridge regression protocol, receiving only the sensor modalities they were designed for. Values report validation R^2 (higher is better); bold marks the best score in each column.

	Sorghum					Maize			
Frozen Encoder	Height	Leaf Cnt	Branch	Roll	Leaf Len	Height	Leaf Angle	Curl	Stem Rad
DINOv2 ViT-B/14 (RGB) [32]	0.924	0.928	0.840	0.573	0.235	0.561	0.816	0.072	0.588
MultiMAE-B (RGB+D) [3]	0.922	0.925	0.833	0.679	0.232	0.570	0.783	0.069	0.442
Point-MAE (PC) [34]	0.958	0.963	0.863	0.716	0.188	0.509	0.863	0.199	0.426
EmbodiedMAE-B (RGB+D+PC) [10]	0.976	0.979	0.871	0.765	0.265	0.528	0.823	0.139	0.428
CROPCODE Headline (1,000 ep)	**0.983**	**0.988**	**0.912**	**0.798**	**0.371**	**0.747**	**0.967**	**0.739**	**0.841**

Table 4. **Controlled ablations under a shared compute budget (197,400 optimizer steps). (a) Modality value-add:** Comparison across active input stream combinations, where θ denotes procedural parameter tokens included in the shared Dirichlet visibility budget. **(b) Dataset scale and encoder capacity:** Under fixed gradient steps, larger backbones do not improve representation quality, while increasing plant diversity steadily reduces reconstruction error. Validation Chamfer ($\times 10^{-3}$) and linear-probe R^2 are shown.

(a) Modality value-add

	Chamfer	Frozen-Probe R^2 (Validation Set)					
Stream Set	$\times 10^{-3}$	Height	Leaf Cnt	Biomass	Branch	Roll	Leaf Len
Sorghum							
PC	5.00	0.895	0.896	0.892	0.792	0.393	0.088
PC + D	2.45	0.976	0.979	0.982	0.903	0.798	0.307
PC + RGB	2.57	**0.979**	**0.983**	**0.986**	**0.934**	0.808	0.267
PC + RGB + D	**1.85**	0.977	0.980	0.982	0.926	0.781	**0.309**
PC + θ	4.84	0.860	0.864	0.863	0.780	0.481	0.078
PC + D + θ	4.08	0.907	0.909	0.916	0.856	0.694	0.191
PC + RGB + θ	3.00	0.962	0.964	0.967	0.890	**0.810**	0.225
PC + RGB + D + θ	2.53	0.962	0.964	0.965	0.880	0.760	0.249
		Height	Leaf Cnt	Biomass	Branch	Curl	Stem Rad
Maize							
PC	2.63	0.171	0.670	0.610	0.756	0.114	0.289
PC + D	2.55	0.453	0.773	0.683	0.804	0.082	0.550
PC + RGB	1.61	0.620	**0.914**	**0.783**	**0.927**	0.334	0.753
PC + RGB + D	1.21	0.633	**0.914**	0.774	0.926	0.236	0.760
PC + θ	2.49	0.256	0.740	0.662	0.787	0.114	0.363
PC + D + θ	2.54	0.382	0.801	0.670	0.795	0.136	0.408
PC + RGB + θ	1.21	0.509	0.881	0.780	0.912	**0.371**	**0.763**
PC + RGB + D + θ	**0.99**	**0.635**	0.908	0.781	0.919	0.304	0.743

(b) Training dataset scale and encoder capacity

Dataset / Capacity Variant	Epochs	Chamfer	Height	Roll	Leaf Len
Sorghum: Training Dataset Scale (Base Model)					
1,000 Plants	6,169	3.13	0.952	0.716	0.249
3,000 Plants	2,100	2.86	0.932	0.679	0.213
10,000 Plants	631	2.48	0.962	0.749	0.245
10,500 Plants	600	2.53	0.962	0.760	0.249
Sorghum: Encoder Capacity (Full Dataset)					
ViT-Small (25 M)	600	2.86	0.964	**0.817**	**0.277**
ViT-Base (114 M)	600	**2.53**	0.962	0.760	0.249
ViT-Large (332 M)	600	2.68	0.948	0.711	0.246
		Chamfer	Height	Branch	Stem Rad
Maize: Training Dataset Scale (Base Model)					
1,000 Plants	6,169	3.54	0.393	0.851	0.593
3,000 Plants	2,100	1.46	0.524	0.867	0.624
10,000 Plants	631	0.87	0.427	0.853	0.717
10,500 Plants	600	0.99	0.635	0.919	0.743
Maize: Encoder Capacity (Full Dataset)					
ViT-Small (25 M)	600	1.22	0.563	0.886	0.682
ViT-Base (114 M)	600	**0.99**	**0.635**	**0.919**	**0.743**
ViT-Large (332 M)	600	1.06	0.580	0.887	0.614

camera for the headline model). However, applying a probe trained on 100% complete clouds fails completely ($R^2 \rightarrow -0.26$; Fig. 9, Table 6). Partial visibility therefore induces a systematic covariate shift in the latent space, meaning field readouts must be calibrated on sensor-matched coverage.

Morphological complexity. Following distillation, single-image reconstruction error remains invariant across leaf-count deciles (Spearman $\rho = -0.07$ in sorghum and $+0.04$ in maize; Fig. 10), showing that CROPCODE scales gracefully to complex canopies.

4.4. Regime III: Neighbor Occlusion in Realistic Crop Rows

Table 7 (Appendix A) compares reconstruction accuracy on identical test plants evaluated in isolation versus inside densely planted crop rows. Models trained solely on isolated plants break down under row occlusion, with Chamfer error jumping sixteenfold in maize ($1.8 \rightarrow 29.0 \times 10^{-4}\,\mathrm{m}^2$) and tenfold in sorghum ($13.1 \rightarrow 133.8 \times 10^{-4}\,\mathrm{m}^2$). Our echo test confirms that these models mistakenly reconstruct neighboring foliage as part of the target plant. Fine-tuning on simulated rows eliminates 92% of the neighbor-induced error in maize ($29.0 \rightarrow 4.0$) and 85% in sorghum ($133.8 \rightarrow 30.7$), cleanly isolating the target plant behind its neighbors (Fig. 5). Procedural parameter accuracy improves in tandem: row fine-tuning reduces leaf parameter MAE on occluded rows from 0.093 to 0.052 in maize and from 0.096 to 0.061 in sorghum. In maize, whose leaves align roughly parallel to the row, error peaks in broadside views and drops in end-on views, whereas sorghum error is invariant to row yaw due to its random leaf azimuths (Fig. 11).

4.5. Any-Subset Modality Reconstruction and Distillation

Table 8 (Appendix A) summarizes cross-modal generation performance before and after teacher–student distillation, illustrated qualitatively in Figs. 4 and 12. Distillation reduces overall cross-modal generation loss by 30.2% in sorghum and 43.5% in maize. Gains are largest on the hardest single-sensor conversions: in sorghum, RGB-to-parameter MAE drops by 76% ($0.058 \rightarrow 0.014$), point-cloud-to-parameter MAE drops in half ($0.040 \rightarrow 0.020$), and RGB-to-cloud

Chamfer improves by 61% ($2.27 \rightarrow 0.88 \times 10^{-3}$). Combining a visual sensor with procedural parameters yields the lowest geometric errors (0.36–0.44×10^{-3}), and a single unified generalist student trained across all input pairs performs within 0.03 of dedicated two-stream specialists in both species.

4.6. Influence of Procedural Parameter Stream Placement

Tables 4 and 9 (Appendix A) examine how active input modalities and parameter-stream visibility shape the learned encoder representation. Visual streams provide the primary morphological signal: adding RGB or depth to point clouds lifts sorghum height R^2 from 0.895 to 0.979 and 0.976, while adding RGB lifts maize height R^2 from 0.171 to 0.620. By contrast, although depth improves Chamfer distance by 25–28%, it yields almost no gain in frozen-probe R^2.

Exposing procedural parameter tokens to the encoder degrades sensor representations in both species. To separate token-budget dilution from shortcut learning, Table 9 holds the visible sensor token budget fixed at 117.0 tokens while varying parameter visibility. In sorghum, adding the parameter stream degrades every probed trait regardless of visibility level—even when used strictly as a decoder-only reconstruction target (R^2 drops from 0.977 to 0.966 on height). In maize, however, gating parameter visibility to 20% or using parameters purely as a decoder target improves all eleven traits (lifting height R^2 from 0.633 to 0.776). We attribute this divergence to shortcut learning when parameters are visible to the encoder, combined with the structural rank of each generator's parameter space: sorghum parameters are tightly coupled (effective rank two), pulling decoder gradients toward global size, whereas maize spans eleven independent morphological axes that act as rich multi-task supervision (Figs. 13 and 14).

4.7. Divergence Between Chamfer Distance and Probing Metrics

Across twelve architectural variants, Chamfer distance and frozen-probe R^2 correlate broadly (Spearman $\rho = -0.57$ to -0.88) but disagree on critical design choices (Fig. 15). Adding depth cuts Chamfer by 25–28% without improving trait probing, and adding parameter tokens improves maize Chamfer while degrading sorghum representations. Because crop leaves are thin, high-aspect-ratio sheets, clustering predicted points along the central stem yields low nearest-neighbor distance even when whole blades are missing: precision exceeds recall across every model variant (Fig. 16). Our loss ablation (Appendix G.13) confirms this mechanism: QAL alone places points precisely on stems but achieves low recall, Sinkhorn divergence alone maximizes coverage with poor precision, and combining both (Eq. 4) yields the best F_1 across thresholds. We therefore rely on frozen-latent probes to guide modality decisions while training with the combined QAL + Sinkhorn loss.

4.8. Real-World Field Validation

We next evaluate our row models on crop rows constructed from real-world plants: procedural reconstructions of 520 real maize plants from MaizeField3D [18, 25] and 30 laser-scanned real sorghum plants (2.2–5.3 m tall). Plants are arranged into rows at 15–25 cm spacing in two orientations (leaf planes parallel and perpendicular to the row) and imaged across 18 rover camera stops per row (9,360 stops per layout for maize; 540 for sorghum), retaining synthetic rendering to isolate morphological sim-to-real transfer from sensor noise.

Evaluated zero-shot, the row fine-tuned model outperforms both the undistilled headline model and the isolated control across both maize row layouts (Table 10, Appendix A), confirming that row-occlusion reasoning transfers to real plant geometry. However, absolute errors remain roughly an order of magnitude higher than on synthetic rows due to a size mismatch: 83% of the real maize plants are taller than the tallest synthetic training plant (1.33 m), capping zero-shot predictions around 1.24 m (median height error is 14% on in-range plants vs. 38% on taller plants). Similarly, real sorghum plants (up to 5.3 m) far exceed the synthetic training range (1.74 m), preventing zero-shot transfer.

Fine-tuning the row model for 15 epochs on 75% of the real plants (split strictly by plant) closes most of this gap on the held-out 25% (130 maize plants and 8 sorghum plants): median maize height error drops from 36% to 1.6%, leaf count accuracy within ± 1 rises from 45% to 70%, maize Chamfer falls from 59.5 to $19.7 \times 10^{-4}\,\mathrm{m}^2$, and sorghum Chamfer drops from 2,113 to $79.4 \times 10^{-4}\,\mathrm{m}^2$. These improvements hold consistently across single- and multi-sensor inputs, row layouts, and occlusion levels (Table 11; Figs. 18, 19, and 17), requiring only 10.6 ms per camera stop at inference.

4.9. Representation Quality, Scaling, and Transfer

Comparison against published encoders. Table 3 benchmarks CROPCODE against DINOv2, MultiMAE, Point-MAE, and EmbodiedMAE using identical mean-pooled token readouts and only the input modalities each encoder supports (Appendix D). Three-stream CROPCODE (PC+RGB+D) outperforms all baselines on 8 of 9 sorghum traits and 10 of 11 maize traits, while our 1,000-epoch headline model achieves the strongest overall representation in the table, beating every published encoder on all eleven maize traits. Notably, full-pose Point-MAE appears artificially strong on synthetic maize due to a renderer artifact in which canonical leaf alignments leak azimuth through camera pose matrices; under realistic gravity-aligned poses,

Point-MAE height R^2 drops from 0.720 to 0.588, whereas pose-free CROPCODE is unaffected (Appendix D). Masked reconstruction comparisons are reported in Appendix G.9.

Scaling, cross-species transfer, and latent geometry. Increasing training dataset size from 1,000 to 10,500 plants monotonically reduces Chamfer distance ($3.13 \rightarrow 2.53 \times 10^{-3}$ in sorghum; $3.54 \rightarrow 0.99 \times 10^{-3}$ in maize), whereas maize probe scores fluctuate at intermediate scales before peaking at the full dataset. Under a fixed gradient-step budget, scaling backbone capacity from ViT-Base (114 M) to ViT-Large (332 M) degrades representation quality in both species (Table 4, Fig. 20). Pretraining on maize and fine-tuning for 150 epochs on sorghum outperforms training from scratch across every probed trait (validation Chamfer 1.83 vs. 3.38×10^{-3}), although decoders do not transfer zero-shot across species. Finally, t-SNE projections of frozen `[CLS]` tokens reveal smooth, continuous manifolds organized by plant scale and leaf posture rather than discrete clusters (Fig. 21), with low effective participation ratios (4.4–6.3 of 768 dimensions in sorghum; 6.7–11.3 in maize).

5. Discussion and Limitations

We summarize the main takeaways and limitations of our study, with a fuller discussion in Appendix E.

Design rule for privileged generative parameters. Because procedural parameter tokens state the target morphological traits directly yet are unavailable at field deployment, allowing the encoder to observe them creates a shortcut that starves visual and point-cloud encoders of gradient signal. This mirrors modality competition [35] and negative gradient transfer [50]. Our budget-matched controls (Table 9) demonstrate that privileged generative parameters should never be largely visible to the encoder during pretraining; when restricted strictly to a decoder target, their benefit depends on whether the generator's parameter space spans independent traits (helping maize) or collapses onto a low-rank axis (hurting sorghum).

Evaluating slender 3D geometry. Nearest-neighbor metrics such as Chamfer distance reward points clustered along thick central stems while overlooking missing leaf blades. Because Chamfer distance and frozen-latent trait probes diverge on depth inclusion, dataset scaling, and partial visibility (Sec. 4.7), geometric metrics alone are insufficient for thin botanical structures. Pairing distribution-matching training objectives (Eq. 4) with downstream trait probes provides a much more reliable signal.

Partial sensing and row occlusion. Camera elevation is the primary driver of single-view difficulty ($\eta^2 = 0.24$–0.31). Under partial point-cloud coverage, morphological information remains preserved in the latent space but undergoes a covariate shift, requiring linear readouts to be refit on sensor-matched inputs. In dense rows, training with physically simulated contact occlusion enables the network to separate the target plant from interleaved neighbors, eliminating 85–92% of neighbor-induced error.

Limitations. First, controlled ablations were run with a single random seed per arm under a fixed step budget (run-to-run variation is roughly 2% in Chamfer and ≤ 0.011 in probe R^2). Second, inverse procedural reconstruction is inherently bounded by the expressivity of the underlying generator G: morphological anomalies absent from G—such as severe disease lesions, broken stalks, or unmodeled tassel droop—are projected onto the nearest valid procedural configuration. Third, synthetic rendering pipelines can introduce subtle pose regularities, underscoring the importance of real-world validation (Sec. 4.8).

6. Conclusion

CROPCODE shows that sparse, heavily occluded field observations are sufficient to recover complete, editable procedural plant models when the procedural generator serves as both the training data engine and the structured output space. By combining four-stream masked pretraining, cross-modal teacher–student distillation, and physically simulated row-occlusion fine-tuning, our framework converts any subset of RGB, depth, and point-cloud inputs into biologically interpretable crop models in milliseconds. Distillation cuts cross-modal generation loss by 30–44%, row fine-tuning eliminates 85–92% of neighbor-induced occlusion error, and brief fine-tuning on real plants brings held-out maize height error down to 1.6%. Beyond digital agriculture, our experiments establish two general principles for structured 3D learning: privileged generative parameters should be withheld from the encoder during multimodal pretraining, and downstream trait probes should accompany nearest-neighbor geometric metrics whenever evaluating slender, occluded objects. We will release FIELDSCAN—comprising 30,000 synthetic plants across two species with multimodal captures, simulated row occlusion, exact procedural ground truth, and a real-field evaluation split—to support further research in 3D plant phenotyping and inverse procedural modeling.

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A. Result Tables

This appendix provides the full numerical tables referenced in the main paper: single-view isolated plant reconstruction (Table 5), partial camera coverage (Table 6), simulated row occlusion (Table 7), any-subset cross-modal generation and distillation (Table 8), token-budget control ablations (Table 9), and real-plant row experiments (Tables 10 and 11).

B. Result Figures

The result figures referenced in the main text are grouped below in order of appearance.

C. Evaluation Metrics

Geometric accuracy is measured by symmetric squared Chamfer distance in the unit-sphere normalized frame ($\times 10^{-3}$; a value of 1.0 corresponds to roughly 33 mm RMS error on a median sorghum plant and 19 mm on a median maize plant, computed using per-plant metric scale factors). Pretrained models are evaluated under their 80% masked pretraining regime, whereas distilled students are evaluated with unmasked inputs.

Procedural parameter accuracy is measured as mean absolute error (MAE) across normalized $[0, 1]$ fields of masked, non-padding leaf tokens, and is also expressed as normalized trait *skill*:

$$\text{Skill} = 1 - \frac{\text{MAE}}{\text{MAE}_{\text{mean}}}, \tag{6}$$

where MAE_{mean} is the error of predicting the training-set mean, so $\text{Skill} = 1$ denotes perfect recovery and $\text{Skill} = 0$ corresponds to predicting the dataset mean. Representation quality is evaluated via ridge regression R^2 on frozen encoder features with 95% paired-bootstrap confidence intervals across 2,250 validation and 2,250 test plants. Because sorghum's four primary developmental traits have effective rank two (stem length and leaf count correlate at $r = 0.994$), sorghum probes also include independent generator axes (mean leaf roll and blade length), whereas maize spans eleven largely independent morphological traits.

D. Encoder-Comparison Probe Protocol

Standardized probing protocol. Table 3 evaluates all encoders under an identical linear probing protocol adapted so every baseline can participate fairly. First, each plant is represented by mean-pooling its spatial/point output tokens (excluding the `[CLS]` token in CROPCODE), which every encoder supports. Second, each model receives only the input modalities it was pretrained on from view 00: DINOv2 receives RGB, MultiMAE receives RGB and depth, Point-MAE receives the point cloud, EmbodiedMAE receives all three sensor streams, and CROPCODE receives its pretrained sensor streams (procedural parameter inputs are always excluded). For every encoder and trait, token features across the 10,500 training plants are standardized using training-split statistics alone, and a ridge regressor is fit with regularization strength $\alpha \in \{0.1, 1, 10, 10^2, 10^3, 10^4\}$ chosen via five-fold cross-validation. We report R^2 on the 2,250 validation plants; an end-to-end supervised baseline trained directly on trait labels is probed in the exact same way on its frozen penultimate features. Probed ground-truth traits are stem length, leaf count, mean branching angle, mean leaf roll, and mean leaf length in sorghum, and summed internode length, mean leaf angle, mean leaf curvature, and stem radius in maize. Because validation and test splits are enriched with extreme phenotypes ($1.8\times$ training variance in sorghum and 1.5–$2.6\times$ in maize), these R^2 values provide a strict internal ranking across encoders rather than absolute scores comparable to unenriched datasets. Test R^2 closely tracks validation (e.g., maize headline height reaches 0.747 on validation and 0.759 on test).

Camera pose artifact analysis. When Point-MAE is supplied with full 3D camera pose matrices on synthetic maize, its apparent accuracy is inflated by a rendering regularity: synthetic maize models align leaf planes along a canonical world axis, allowing full pose matrices to leak plant azimuth. Evaluating Point-MAE with realistic gravity-aligned camera poses (where azimuth is uncalibrated) reduces its height R^2 from 0.720 to 0.588, whereas pose-free CROPCODE is unaffected.

E. Extended Discussion

We expand below on the architectural, methodological, and practical findings from our experiments.

Privileged parameter visibility and modality competition. A central question when pretraining multimodal models alongside procedural generators is how ground-truth structural parameters should be integrated. The parameter stream θ states the target phenotypic traits explicitly, yet it is never observed at deployment. Across our ablations (Sec. 4.6), allowing the encoder to attend to parameter tokens consistently degraded sensor representation quality in both species (Table 9).

This behavior aligns with shortcut learning, modality competition [35], and negative gradient transfer [50]. Because parameter tokens directly encode the target geometry, the encoder learns a low-cost shortcut through them, starving the RGB, depth, and point-cloud encoders of informative gradient updates. When parameter tokens are removed at inference, the resulting sensor representations underperform.

Our budget-matched controls (Table 9) further reveal how decoder-only parameter supervision interacts with generator structure:

- **Sorghum:** Keeping parameters strictly as a decoder target ($\rho_\theta = 1.0$) still reduces sensor probe R^2 relative to a three-stream model (height drops from 0.977 to 0.966).

Table 5. **Regime I: Single-view reconstruction on isolated plants.** Each row evaluates the distilled generalist given a single input stream on view 00 across all 2,250 validation plants per species. †The undistilled headline model given all three sensor streams is evaluated under its 80% masked pretraining regime (passing 100% visible tokens causes cloud collapse). Leaf count and height are decoded directly from the predicted procedural recipe: maize stores leaf count explicitly in the plant token, whereas sorghum derives leaf count from predicted stem length via the generator relation (stem length $= 0.05n - 0.001$). Leaf-parameter skill is Eq. (6) averaged across active leaf slots. Chamfer distance ($\times 10^{-3}$) and F_1 score are computed against the complete unit-sphere plant cloud (note that Table 8 averages across random views, explaining the slight numerical difference). The three-view row averages predicted recipes across views 00, 01, and 02. The two-stage baseline generates a point cloud from view 00 RGB and then predicts the recipe from that generated cloud; reading recipes from generated clouds loses nearly all parameter accuracy (sorghum leaf skill drops from 0.656 to -0.010), confirming the advantage of direct end-to-end parameter inference. RMS error converts unit-sphere Chamfer into millimeters per plant (median plant scale: 1.03 m in sorghum, 0.61 m in maize).

Input Modality	Leaf Count (Acc. ± 1) ↑	Height Error (Rel. Err.) ↓	Leaf Parameters (Mean Skill) ↑	F_1-score ($\tau = 0.01$) ↑	Chamfer $\times 10^{-3}$ ↓	RMS Error (mm) ↓
Sorghum						
RGB	0.909	0.036	0.656	0.386	1.36	37.1
Depth	0.956	0.026	0.679	**0.440**	**0.93**	**30.3**
Point Cloud	**0.998**	0.022	0.675	0.370	1.45	38.7
RGB + D + PC (headline)†	0.889	0.038	0.636	0.392	1.36	37.0
RGB, three views (recipes averaged)	0.995	**0.018**	**0.747**	–	–	–
Two-stage: RGB → cloud → recipe	0.301	0.221	−0.010	0.386	1.36	37.1
Maize						
RGB	0.973	0.045	0.567	0.482	0.83	13.6
Depth	0.972	0.046	0.571	**0.529**	**0.64**	**12.4**
Point Cloud	0.986	**0.043**	0.587	0.455	0.87	15.0
RGB + D + PC (headline)†	0.953	0.045	0.552	0.445	0.93	15.4
RGB, three views (recipes averaged)	**0.988**	0.044	**0.597**	–	–	–
Two-stage: RGB → cloud → recipe	0.066	0.131	0.015	0.482	0.83	13.6

Table 6. **Self-occlusion under partial camera coverage (sorghum test set).** Evaluated on point clouds restricted via mesh ray-casting to points visible from 1 or 3 cameras. *Matched*: linear probe refit per visibility condition. *Transferred*: linear probe fit on 100% complete clouds applied without recalibration. Chamfer distance is evaluated against the complete ground-truth cloud. Because maize point clouds are single-view captures, the maize model is trained on the 1-camera condition, its transferred probe is fit there and evaluated on multi-view fused clouds, and Chamfer is measured against the 10-camera validation cloud.

	Matched Probe (R^2, Height)			Transferred Probe (R^2, Height)			Chamfer ($\times 10^{-3}$)		
Input Modality	Full	3-Cam	1-Cam	Full	3-Cam	1-Cam	Full	3-Cam	1-Cam
Cloud Coverage	100%	78%	49%	100%	78%	49%	100%	78%	49%
Point Cloud (PC)	0.888	0.839	0.765	0.888	0.509	0.575	4.99	6.18	12.8
PC + RGB	0.977	0.967	0.948	0.977	0.624	0.001	2.49	3.64	11.5
PC + RGB + D	0.975	0.968	0.956	0.975	0.721	0.576	1.76	3.06	11.4
Four-Stream Model	0.961	0.953	0.946	0.961	0.748	0.818	2.52	3.62	11.4
Headline Teacher	**0.982**	**0.975**	**0.960**	**0.982**	0.623	−0.26	**1.36**	3.14	11.6
Maize: Full = 10 cameras fused; coverage 100 / 96 / 72% of the 10-camera cloud									
Headline Teacher	0.719	0.698	0.754	0.108	0.251	0.754	4.98	4.54	1.15

Because sorghum parameters have an effective rank of two, decoder gradients pull the shared latent space toward global plant scale (PC1 accounts for 44% of latent variance and correlates with leaf count at $|r| = 0.95$, reducing participation ratio from 6.7 to 4.3).

- **Maize:** Keeping parameters as a decoder target ($\rho_\theta = 1.0$) or gating them to 20% visibility improves probe R^2 across all eleven traits (lifting height R^2 from 0.633 to 0.776). Because maize spans eleven independent morphological axes, reconstructing parameters provides rich, multi-task structural supervision without inducing encoder shortcuts.

Together, these results yield a clear design rule: *privileged generative parameters should never be largely visible to the encoder during pretraining*. Restricting them to a decoder reconstruction target (or heavy gating $\rho_\theta \leq 0.20$) eliminates encoder shortcuts while retaining generative parameter prediction.

Metric divergence on slender geometry. Standard Chamfer distance measures average nearest-neighbor surface displacement:

$$\mathrm{CD}(P, \hat{P}) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in \hat{P}} \|p - q\|_2^2 + \frac{1}{|\hat{P}|} \sum_{q \in \hat{P}} \min_{p \in P} \|p - q\|_2^2. \tag{7}$$

On thin, high-aspect-ratio leaves, a network can achieve low Chamfer error simply by concentrating points along

Table 7. **Regime III: Reconstruction of identical plants in isolation versus simulated crop rows.** Evaluated across 2,250 held-out test plants per species over 10 row orientations ($N = 22{,}500$). Chamfer distance is reported in metric units (10^{-4} m^2). Δ reports the paired within-plant difference (row minus isolated) with 95% bootstrap confidence intervals over plants.

		Chamfer vs. Target (10^{-4} m^2)			Leaf Parameter MAE		
Model Variant	Sensor Input	Isolated	Row	Δ [95% CI]	Isolated	Row	Δ [95% CI]
Maize: Zero-Shot (Headline Model)							
	RGB	56.7	59.7	$+3.0$ [$+2.4$, $+3.6$]	0.085	0.088	$+0.003$ [$+0.003$, $+0.003$]
	Depth	47.2	62.6	$+15.3$ [$+14.8$, $+15.9$]	0.090	0.093	$+0.002$ [$+0.002$, $+0.003$]
	Point Cloud	20.0	33.8	$+13.7$ [$+13.4$, $+14.1$]	0.064	0.081	$+0.018$ [$+0.017$, $+0.018$]
	RGB + D + PC	26.4	29.6	$+3.2$ [$+2.9$, $+3.6$]	0.086	0.093	$+0.007$ [$+0.007$, $+0.008$]
Maize: Row Fine-Tuned							
	RGB	13.7	14.4	$+0.6$ [$+0.4$, $+0.8$]	0.067	0.071	$+0.004$ [$+0.003$, $+0.004$]
	Depth	10.3	9.6	-0.7 [-0.9, -0.5]	0.061	0.064	$+0.003$ [$+0.003$, $+0.004$]
	Point Cloud	21.7	21.9	$+0.2$ [-0.1, $+0.5$]	0.063	0.073	$+0.010$ [$+0.010$, $+0.010$]
	RGB + D + PC	12.6	**4.0**	-8.6 [-8.8, -8.3]	**0.046**	**0.052**	$+0.005$ [$+0.005$, $+0.005$]
Maize: Isolated Control							
	RGB	11.6	56.1	$+44.5$ [$+44.0$, $+45.1$]	0.071	0.095	$+0.024$ [$+0.023$, $+0.024$]
	Depth	4.6	52.8	$+48.2$ [$+47.5$, $+48.9$]	0.059	0.097	$+0.038$ [$+0.037$, $+0.039$]
	Point Cloud	8.2	25.1	$+16.9$ [$+16.6$, $+17.3$]	0.046	0.076	$+0.030$ [$+0.029$, $+0.030$]
	RGB + D + PC	1.8	29.0	$+27.2$ [$+26.9$, $+27.6$]	0.043	0.093	$+0.050$ [$+0.049$, $+0.051$]
Sorghum: Zero-Shot (Headline Model)							
	RGB	163.1	218.5	$+55.4$ [$+52.6$, $+58.4$]	0.129	0.135	$+0.006$ [$+0.006$, $+0.006$]
	Depth	106.2	207.2	$+101.0$ [$+98.0$, $+104.2$]	0.070	0.109	$+0.039$ [$+0.039$, $+0.040$]
	Point Cloud	177.7	231.4	$+53.8$ [$+52.2$, $+55.4$]	0.114	0.127	$+0.013$ [$+0.012$, $+0.013$]
	RGB + D + PC	174.3	230.5	$+56.2$ [$+53.9$, $+58.5$]	0.108	0.114	$+0.006$ [$+0.006$, $+0.006$]
Sorghum: Row Fine-Tuned							
	RGB	34.8	29.8	-5.0 [-5.4, -4.7]	0.094	0.097	$+0.003$ [$+0.003$, $+0.004$]
	Depth	24.4	18.6	-5.8 [-6.1, -5.5]	0.078	0.055	-0.022 [-0.023, -0.022]
	Point Cloud	46.0	70.8	$+24.8$ [$+24.4$, $+25.2$]	0.074	0.082	$+0.007$ [$+0.007$, $+0.007$]
	RGB + D + PC	38.7	**30.7**	-8.0 [-8.5, -7.6]	**0.073**	**0.061**	-0.012 [-0.013, -0.011]
Sorghum: Isolated Control							
	RGB	15.7	133.2	$+117.5$ [$+115.8$, $+119.3$]	0.094	0.114	$+0.020$ [$+0.020$, $+0.021$]
	Depth	6.6	171.9	$+165.3$ [$+162.9$, $+167.8$]	0.042	0.100	$+0.059$ [$+0.058$, $+0.060$]
	Point Cloud	30.9	100.9	$+70.0$ [$+68.8$, $+71.3$]	0.065	0.083	$+0.018$ [$+0.018$, $+0.019$]
	RGB + D + PC	13.1	133.8	$+120.8$ [$+119.5$, $+122.1$]	0.049	0.096	$+0.047$ [$+0.046$, $+0.048$]

Table 8. **Any-subset cross-modal reconstruction before and after distillation (validation set).** Each row evaluates reconstruction of the missing streams given only the listed source modality. *Before*: zero-shot pretrained initialization; *After*: distilled generalist student model.

	Sorghum (Validation Set)				Maize (Validation Set)			
	θ MAE ↓		Chamfer $\times 10^{-3}$ ↓		θ MAE ↓		Chamfer $\times 10^{-3}$ ↓	
Input Modality	Before	After	Before	After	Before	After	Before	After
RGB Only	0.058	0.014	2.27	0.88	0.037	0.019	11.3	0.77
Depth Only	0.017	**0.013**	0.90	0.63	0.024	0.019	1.26	0.58
Point Cloud Only	0.040	0.020	5.48	1.39	0.020	0.018	6.43	0.83
θ (Parameters Only)	–	–	22.7	6.83	–	–	80.7	12.8
Two-Stream Input Pairs → Point Cloud Generation (Dedicated Student / Generalist)								
RGB + D	–	–	2.91	0.60 / 0.59	–	–	1.30	0.53 / 0.55
RGB + θ	–	–	0.43	0.40 / 0.39	–	–	0.47	0.44 / 0.41
D + θ	–	–	0.50	0.37 / **0.36**	–	–	0.46	0.42 / **0.40**
Mean Generation Loss	$0.371 \rightarrow 0.259$ (-30.2%)				$0.290 \rightarrow 0.164$ (-43.5%)			

the thicker central stem while omitting entire leaf blades. Frozen-latent trait probes directly test whether the encoder captures full morphological structure. Consequently, the two metrics disagree on key design decisions:

1. **Adding depth:** Depth cuts Chamfer distance by 28% in sorghum and 25% in maize, yet leaves frozen-probe R^2 virtually unchanged (Table 4), indicating that depth refines local surface placement in the decoder without enriching high-level structural features in the encoder.
2. **Dataset scaling:** Scaling from 1,000 to 10,000 sorghum plants lowers Chamfer by 21% ($3.13 \rightarrow 2.48 \times 10^{-3}$), whereas probe scores saturate (Table 4).
3. **Partial visibility:** Single-camera clouds increase Chamfer error by $2.6\times$ to $6.5\times$, while refit probes on fused multimodal inputs lose at most 0.030 in height R^2 (Table 6).

Combining distribution-matching losses (Eq. 4) with downstream trait probes is therefore essential when developing

Table 9. **Token-budget control ablations.** Four-stream models trained with procedural parameter tokens (θ) gated independently of the sensor visibility budget, preserving an identical allocation of 117.0 visible sensor tokens across parameter visibility levels (0%, 20%, 60%, and decoder target-only). Because maize uses 29 parameter tokens versus 25 in sorghum, exact visible fractions differ slightly after rounding to integer token counts (the maize 20% arm keeps 5 of 29 tokens). Validation Chamfer distance ($\times 10^{-3}$) and frozen-probe R^2 are reported.

	Vision	θ Vis.	Chamfer	R^2 (Validation Set)			
Variant	Budget	Ratio	$\times 10^{-3}$	Height	Leaf Cnt	Roll	Leaf Len
Sorghum							
PC + RGB + D	117.0	–	**1.85**	**0.977**	**0.980**	**0.781**	**0.309**
+θ, Shared Budget	107.4	58%	2.53	0.962	0.964	0.760	0.249
+θ, Gated 20%	117.0	20%	2.29	0.964	0.967	0.771	0.278
+θ, Gated 60%	117.0	60%	2.40	0.959	0.963	0.761	0.239
+θ, Target Only	117.0	0%	2.49	0.966	0.970	0.702	0.220
Maize				Height	Leaf Cnt	Curl	Stem Rad
PC + RGB + D	117.0	–	1.21	0.633	0.914	0.236	0.760
+θ, Shared Budget	106.6	56%	0.99	0.635	0.908	0.304	0.743
+θ, Gated 20%	117.0	17%	**0.94**	0.704	0.931	**0.464**	0.791
+θ, Gated 60%	117.0	59%	1.04	0.521	0.851	0.303	0.688
+θ, Target Only	117.0	0%	1.16	**0.776**	**0.947**	0.357	**0.804**

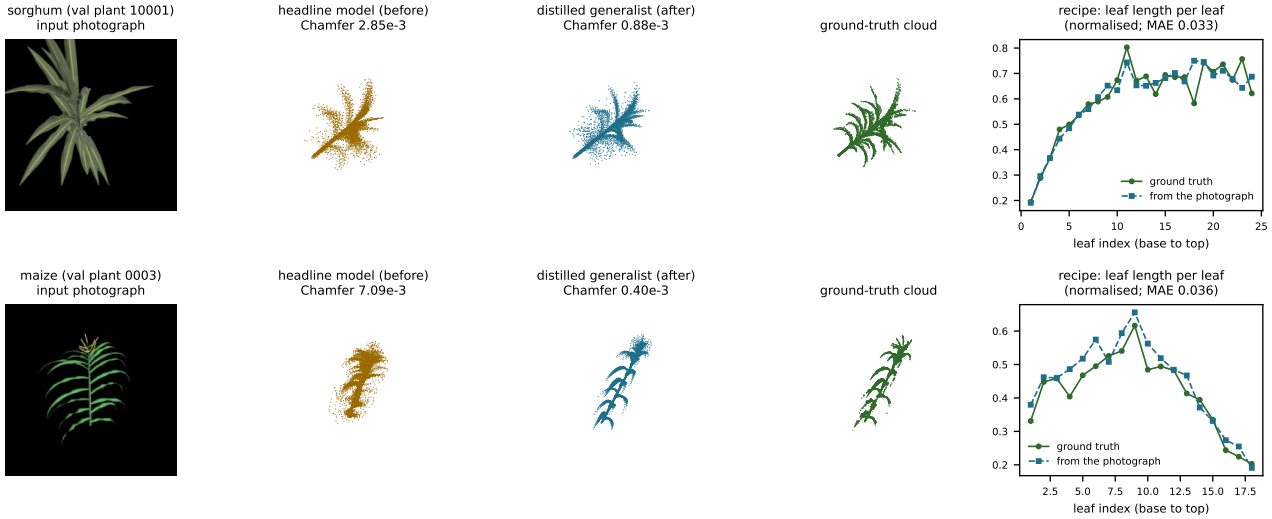


Figure 6. **Single-photograph 3D reconstruction on held-out test plants.** Each row shows inference from a single input RGB image (left). Compared to the undistilled headline model, the distilled generalist resolves individual leaf blades and lowers Chamfer distance (shown above each cloud). The rightmost column compares the generalist's predicted normalized leaf lengths against ground truth from base to top.

models for thin plant geometry.

Partial sensing, covariate shift, and row occlusion. Field observations are inherently incomplete, and our experiments clarify three failure modes and remedies: (1) camera elevation accounts for $\eta^2 = 0.24$–0.31 of single-view error variance because steep top-down or bottom-up views cause leaves to self-occlude along the optical axis; (2) partial cloud coverage preserves latent trait information but shifts feature statistics, causing probes trained on complete clouds to fail ($R^2 \to -0.26$) unless refit on sensor-matched inputs; and (3) fine-tuning on simulated crop rows trains the network to segment out interleaved neighbor foliage, removing 92% (maize) and 85% (sorghum) of neighbor-induced error. Moreover, because $G(\hat{\theta})$ enforces a valid botanical structure, procedural outputs degrade far more gracefully under occlusion than unstructured point clouds.

Capabilities and limits of cross-modal distillation. Distilling a four-stream teacher into single-sensor students lowers cross-modal generation loss by 30.2% in sorghum and 43.5% in maize (Table 8), concentrating gains on weak input modalities (e.g., cutting RGB-to-cloud Chamfer by 61%) and difficult plant geometries (Fig. 7B). Jointly supervising all four output streams also outperforms a parameter-only specialist student (point-cloud recipe MAE of 0.019 vs. 0.036 and 0.038 at learning rates 10^{-4} and 3×10^{-5} from the same warm start at 0.040), showing that auxiliary visual reconstruction regularizes parameter regression. However, distillation cannot compensate for missing camera viewpoints

Table 10. **Real-world plants arranged in crop rows (E10).** All three sensor streams are provided as input. Target Chamfer distance is reported in 10^{-4} m^2 with 95% bootstrap confidence intervals over plants; Miss@5 mm is the fraction of ground-truth target points farther than 5 mm from any predicted point; Height is the median relative height error; and Leaves ± 1 is the share of camera stops whose predicted leaf count falls within one leaf of ground truth (sorghum scans do not include comparable height or leaf-count labels). *Real-row FT* denotes the row fine-tuned model further trained on 75% of the real plants. For reference, on synthetic test rows the row fine-tuned model achieves Chamfer errors of 4.0 (maize) and 30.7 (sorghum).

Rows (plants)	Model	Chamfer ↓	Miss@5 mm ↓	Height ↓	Leaves ± 1 ↑
Zero-shot, all real plants					
Maize parallel (520)	Headline (no row FT)	75.1 [73.1, 77.2]	0.92	53%	17%
	Isolated control	55.6 [54.2, 57.2]	0.93	43%	4%
	Row fine-tune (FT)	**41.2** [39.7, 42.8]	0.93	**37%**	**47%**
Maize perpend. (520)	Headline (no row FT)	121.2 [116.6, 127.2]	0.96	52%	23%
	Isolated control	137.3 [133.6, 141.9]	0.98	40%	13%
	Row fine-tune (FT)	**80.0** [76.4, 85.2]	0.97	**35%**	**46%**
Sorghum, both (30)	Headline (no row FT)	2,541 [2,205, 2,886]	0.99	–	–
	Isolated control	**975** [876, 1,079]	0.99	–	–
	Row fine-tune (FT)	2,145 [1,910, 2,376]	0.99	–	–
Fine-tuned on 75% of the real plants, scored on the held-out 25%					
Maize, both (130)	Row FT	59.5 [56.0, 63.2]	0.95	36%	45%
	Real-row FT	**19.7** [18.6, 21.0]	**0.87**	**1.6%**	**70%**
Sorghum, both (8)	Row FT	2,113 [1,647, 2,520]	0.99	–	–
	Real-row FT	**79.4** [68.3, 92.6]	**0.92**	–	–

Table 11. **Breakdown of real-row fine-tuning performance (E10, held-out plants). Top:** Target Chamfer distance (10^{-4} m^2) across single- and multi-sensor inputs before (Row FT) and after (Real-row FT) fine-tuning on 75% of the real plants. **Bottom:** Maize results (all three streams) stratified by ground-truth plant height, showing median relative height error and Chamfer distance (the procedural recipe's normalized height ceiling is 2.0 m).

	Maize (130 plants)		Sorghum (8 plants)	
Input	Row FT	Real-row FT	Row FT	Real-row FT
Image + depth + cloud	59.5	**19.7**	2,113	**79.4**
Image only	95.4	48.7	1,033	106.2
Depth only	76.8	43.6	834	89.0
Cloud only	75.0	47.4	3,773	311.8

		Height error		Chamfer	
Maize plant height	Plants	Row FT	Real-row FT	Row FT	Real-row FT
$\leq$1.33 m (training range)	22	15%	**2.0%**	47.0	**16.3**
1.33–1.6 m	45	30%	**1.2%**	57.7	**18.8**
1.6–2.0 m	51	41%	**1.4%**	62.7	**20.4**
$>$2.0 m	12	51%	**5.9%**	75.4	**26.7**

during pretraining (Fig. 8), and parameters submitted to inverse regression must be physically observable from field sensors (early experiments with unobservable leaf waviness parameters exhibited high view-dependent variance and near-zero skill, prompting their removal from the final generator).

Limitations and future work. First, controlled ablations were run with one seed per configuration at a fixed 197,400-step budget (estimated run-to-run variation is $\approx 2\%$ in Chamfer and ≤ 0.011 in probe R^2). Second, recovered models are constrained by the expressivity of the procedural generator

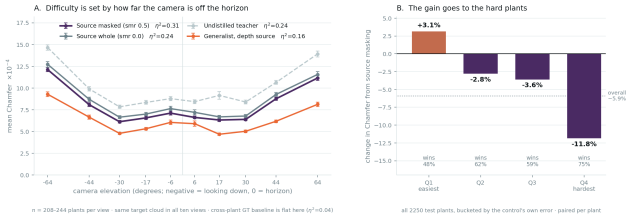


Figure 7. **Camera elevation governs single-view reconstruction difficulty. (A)** Photograph-to-cloud Chamfer distance on test plants across camera elevation angles (208–244 plants per bin). Both pretrained and distilled models follow a symmetric U-shaped curve, performing best on side views and worst on steep top-down or bottom-up views ($\eta^2 = 0.24$–0.31). **(B)** Error reduction from masking 50% of source tokens during distillation across baseline error quartiles, showing the largest gains on difficult plant geometries.

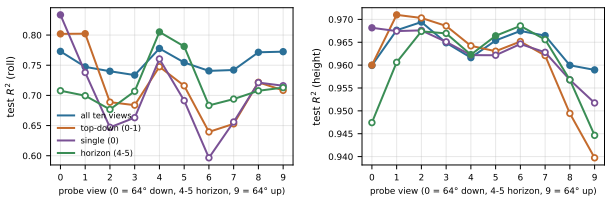


Figure 8. **Impact of training view distribution on downstream trait probing (sorghum test set).** Probes are fit and evaluated per elevation view k for models trained on all ten views versus restricted elevation bands (filled markers denote training elevations). Narrow-band training improves accuracy within the seen elevation range but degrades generalization to unseen camera angles.

G; field anomalies such as severe disease, broken stems, or unmodeled tassel droop cannot be represented exactly and are mapped to the closest valid procedural plant. Expanding procedural grammars to capture stress and damage phenotypes is an important next step. Third, real-field evaluation remains vital to guard against synthetic renderer cues.

F. Detailed Literature Review

This appendix expands upon Section 2 across six related areas.

F.1. Procedural Plant Models and Inverse Procedural Modeling

Procedural plant modeling began with Lindenmayer systems (L-systems) [36] and has evolved into parametric crop generators whose parameters correspond directly to agronomic traits [19, 20]. Early inverse procedural modeling recovered tree generator parameters per instance using stochastic search [44], MCMC inference [45], and L-system induction [17, 28]. Feed-forward inverse approaches have recently emerged for CAD and man-made shapes—including shape

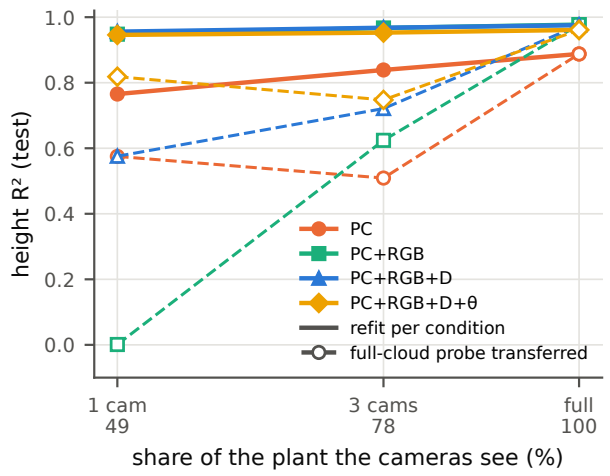


Figure 9. **Persistence of phenotypic information under partial surface coverage.** Test-set height R^2 plotted against visible surface percentage across 1-camera, 3-camera, and full coverage. Refitting the linear probe per visibility level (solid lines) preserves high accuracy, whereas directly transferring a probe trained on complete clouds (dashed lines) fails due to latent covariate shift.

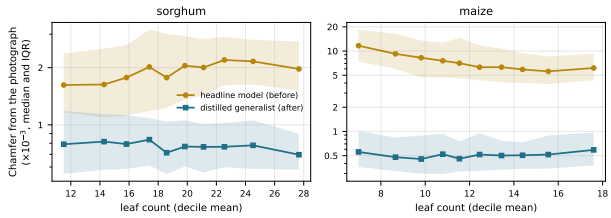


Figure 10. **Reconstruction error across morphological complexity (test set, 2,250 plants per species).** Single-photograph Chamfer distance across leaf-count deciles before and after distillation. After distillation, error remains flat regardless of canopy complexity.

programs [46], diffusion-based inverse procedural generation [55], Text2CAD [24], and cadrille [26]—but they operate on rigid, isolated objects without inter-object occlusion or deformable thin organs.

F.2. Learned Parametric Recovery of Crop Plants

Recent agricultural vision systems infer parametric crop representations from 3D observations. Demeter [8] learns articulated soybean morphology from point clouds, while Ghrer et al. [14] infer grammar-based plant parameters from 3D scans; both assume clean, complete point clouds of isolated plants. CropCraft [54] fits procedural crop parameters via Bayesian optimization against multi-view NeRF depth maps, and Hadadi et al. [9, 18, 19] combine vision-language prompting with differentiable NURBS fitting on terrestrial laser scans, requiring roughly ten minutes per isolated plant. Tree-D Fusion [29] reconstructs urban trees from street-view images using diffusion priors, focusing on woody branch envelopes rather than thin crop foliage.

F.3. Point-Cloud Completion and Slender Geometry

Point-cloud completion networks have progressed from folding decoders [53] to geometry-aware transformers [5, 51, 52], primarily evaluated on compact objects in PCN and ShapeNet-55. While recent methods address amodal completion of occluded plants [27] and thin tubular structures [13, 39], they output unstructured point sets rather than editable parametric models. Moreover, standard Chamfer distance is known to be insensitive to missing thin surfaces [30, 43, 49], motivating our combined use of Sinkhorn divergence and frozen-latent trait probes.

F.4. 3D Plant Phenotyping Datasets

Existing 3D plant datasets such as Pheno4D [42], Crops3D [56], and MaizeField3D [25] provide valuable scans for organ segmentation and trait estimation [2, 37, 38]. However, none pairs real-field captures and physically simulated row occlusion with exact procedural ground-truth parameters across multiple species, a gap FIELDSCAN is designed to fill.

F.5. Multimodal Masked Pretraining and Distillation

Masked autoencoding [21] has been extended to multimodal images [3], any-to-any generation [4, 31], point clouds [34], and RGB–depth–point-cloud inputs in EmbodiedMAE [10]. Cross-modal distillation [22] enables robust inference when sensors are missing at deployment [41]. By integrating continuous numerical parameter tokens [15] into generative cross-modal distillation, we uncover how parameter visibility drives modality competition [35] and gradient interference [7, 50].

F.6. Synthetic Pretraining and Sim-to-Real Transfer

Synthetic pretraining spans fractal patterns [23] to photorealistic procedural scenes [16, 40, 47], with downstream utility governed by task alignment [1, 11]. Because CROPCODE predicts the exact parameter space of the generator and validates transfer on real MaizeField3D captures [25], synthetic pretraining aligns directly with the target task.

G. Detailed Ablation Studies

All ablation models are trained for 197,400 optimizer steps at global batch size 32 and evaluated across all 2,250 validation and 2,250 test plants per species using ridge regression probes on frozen `[CLS]` tokens with 2,000-resample paired bootstrap confidence intervals.

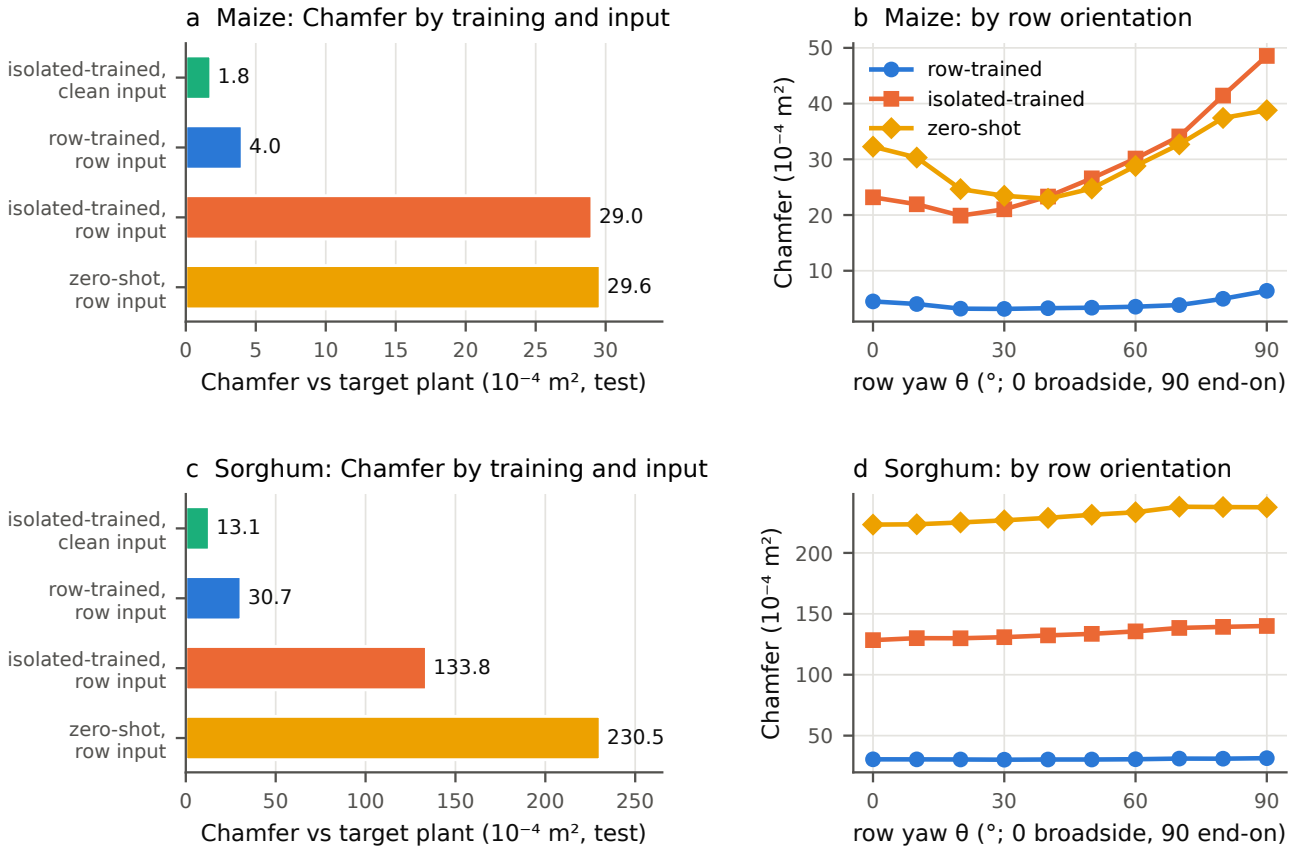


Figure 11. **Quantitative row-occlusion performance on maize (a, b) and sorghum (c, d) test rows.** **(a, c)** Target plant Chamfer distance comparing isolated-trained models on isolated vs. row inputs, row fine-tuned models on row inputs, and zero-shot baselines. **(b, d)** Reconstruction error across row yaw angles (θ). Row fine-tuning eliminates the vast majority of neighbor-induced occlusion error in both species.

G.1. Training View Distribution

We compare pretraining across all ten elevation angles against restricting training to top-down views ($i \in \{0, 1\}$), horizontal views ($i \in \{4, 5\}$), or a single top-down view ($i = 0$). In sorghum (Fig. 8), restricted models improve slightly on their seen elevations (e.g., $+0.061$ roll R^2 on view 0 for the single-view arm) but drop by 0.066–0.144 on unseen angles, whereas the all-view model maintains $R^2 = 0.734$–0.778 across all views. In maize, where evaluation views are sampled randomly on the sphere, every restricted model underperforms the all-view baseline across all traits (validation height R^2 of 0.45, 0.49, and 0.52 for top-down, horizontal, and single-camera arms vs. 0.64 for all views; leaf angle 0.82, 0.80, and 0.89 vs. 0.92; stem radius 0.57, 0.57, and 0.72 vs. 0.75). Notably, the single-camera arm—which samples the full spherical distribution with one-tenth as many images—outperforms the two-view band arms, proving that elevation coverage outweighs raw image count. Point-cloud F_1 at $\tau = 0.01$ confirms this finding (in maize, all-view reaches 0.31 vs. 0.22 horizontal, 0.19 top-down, and 0.17 single-camera).

G.2. Modality Value-Add Ablations (E2)

Table 12 lists validation Chamfer distance and height R^2 across all eight modality combinations under equal compute budgets.

G.3. Token-Budget Control Arms

In standard four-stream pretraining, sensor streams share the 122–123 visible token budget with parameter tokens (leaving 107.4 visible sensor tokens vs. 117.0 in three-stream models). When we decouple parameter visibility to hold visible sensor tokens fixed at 117.0 across parameter visibility rates of 20%, 60%, and 0% (decoder target-only), sorghum still underperforms the three-stream model on every trait (height $R^2 = 0.959$–0.966 vs. 0.977; leaf roll 0.702–0.771 vs. 0.781). Conversely, maize improves across all eleven traits at 20% visibility and 0% target-only supervision.

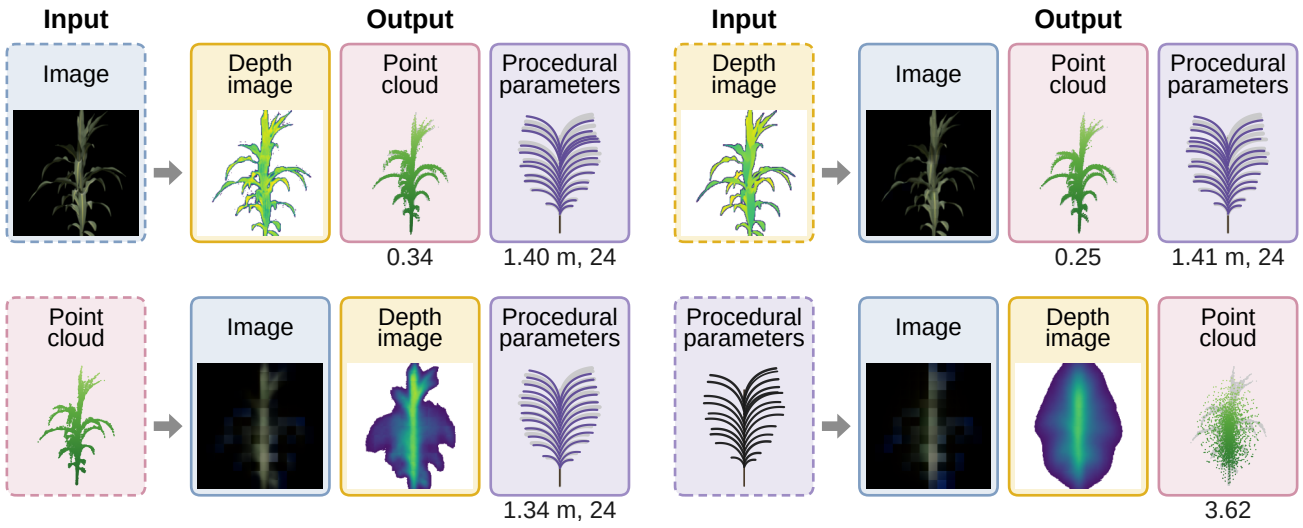


Figure 12. **Any-to-all cross-modal reconstruction on a held-out sorghum test plant.** Format follows Fig. 4 (ground truth: 1.43 m height, 24 leaves). Given any single input modality (dashed box), the distilled generalist accurately reconstructs the remaining three streams.

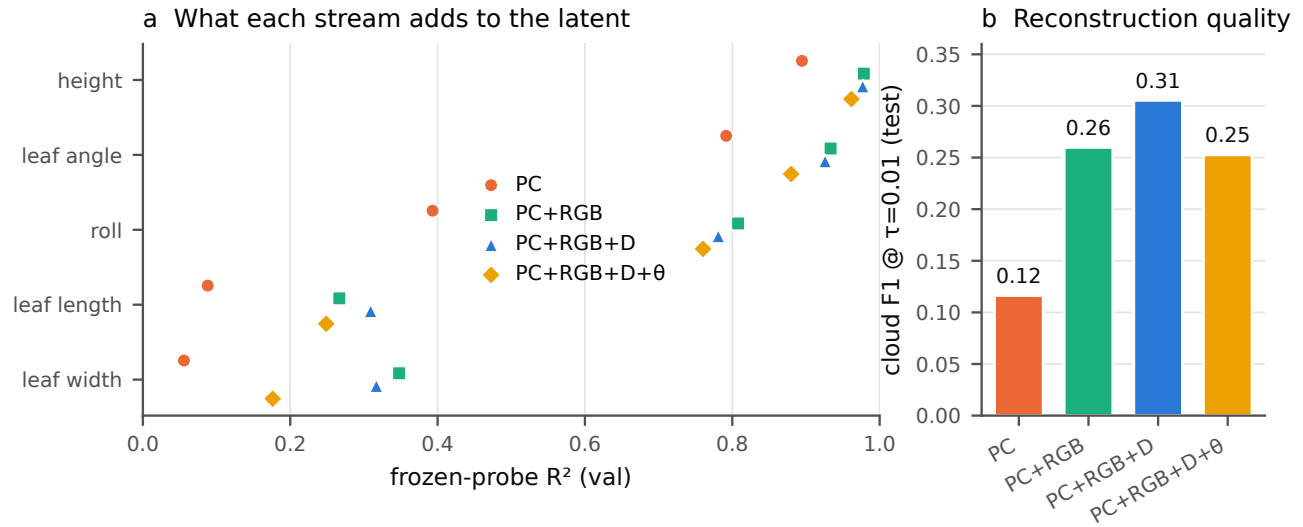


Figure 13. **Modality contributions to latent representations and geometric reconstruction (sorghum). (a)** Frozen-probe R^2 across morphological traits. **(b)** Point-cloud reconstruction F_1 score at threshold $\tau = 0.01$ (≈ 1 cm).

G.4. Parameters as a Decoder Reconstruction Target Only

Supervising procedural parameters strictly as a decoder target ($\rho_\theta = 1.0$) degrades sorghum representations relative to omitting the stream entirely (validation R^2 drops by -0.011 on height, -0.07 on leaf roll, -0.10 on leaf length, and -0.15 on leaf width), concentrating 44% of latent variance into a single size-correlated principal component ($|r| = 0.95$ with leaf count). In maize, target-only parameter supervision achieves the highest overall representation quality across all eleven traits.

G.5. Data Volume and Model Capacity Scaling at Fixed Compute

Table 13 details dataset scaling (1k, 3k, 10k, and 10.5k plants) and encoder scaling (ViT-Small 25 M, ViT-Base 114 M, ViT-Large 332 M) under fixed 197,400-step compute budgets.

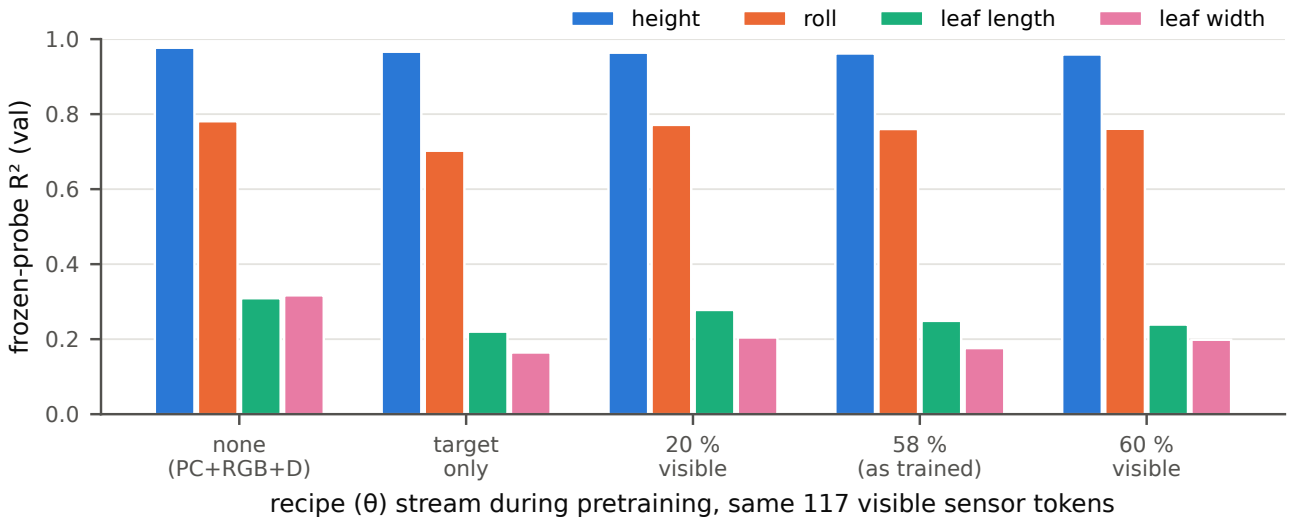


Figure 14. **Effect of procedural parameter visibility during pretraining.** Linear probe R^2 across parameter visibility levels under matched sensor token budgets (117.0 visible sensor tokens).

Table 12. **Extended modality value-add ablations (E2, validation set).** In sorghum, adding parameter tokens (θ) to the encoder degrades both Chamfer error and height R^2 across multimodal combinations. In maize, parameter tokens lower Chamfer error while their impact on probing depends on the accompanying sensor streams.

	Sorghum		Maize	
Experimental Variant	Chamfer ($\times 10^{-3}$)	Height R^2	Chamfer ($\times 10^{-3}$)	Height R^2
Point Cloud (PC)	5.00	0.895	2.63	0.171
PC + Depth	2.45	0.976	2.55	0.453
PC + RGB	2.57	0.979	1.61	0.620
PC + RGB + Depth	1.85	0.977	1.21	0.633
PC + Parameters (θ)	4.84	0.860	2.49	0.256
PC + Depth + θ	4.08	0.907	2.54	0.382
PC + RGB + θ	3.00	0.962	1.21	0.509
Full Four-Stream Model (PC + RGB + Depth + θ)	2.53	0.962	0.99	0.635

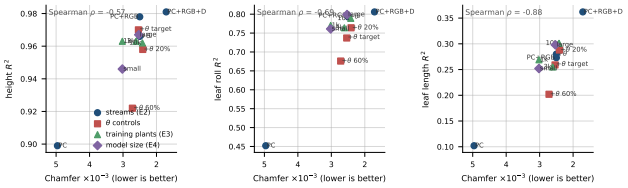


Figure 15. **Divergence between Chamfer distance and frozen-latent trait probes.** Validation Chamfer distance plotted against height probe R^2 across twelve model variants.

G.6. Distillation Dynamics per Source Modality

Cross-modal distillation reduces mean generation loss by 30.2% in sorghum and 43.5% in maize over 100 epochs, with the largest improvements on single-sensor inputs: sorghum point-cloud-to-cloud Chamfer improves from 5.48 to 1.39×10^{-3}, parameter-only-to-cloud Chamfer drops from 22.7 to 6.83×10^{-3}, and maize RGB-to-cloud Chamfer drops from 11.3 to 0.77×10^{-3}. Moreover, on full validation point clouds, our multi-stream distilled generalist achieves a parameter MAE of 0.019 compared to 0.036 and 0.038 for parameter-only specialist students trained from the same initialization (at learning rates 10^{-4} and 3×10^{-5}), demonstrating the regularizing benefit of joint multimodal reconstruction.

G.7. Label-Free View Invariance Across Camera Perspectives

Evaluating prediction consistency across all ten views of each plant provides a label-free measure of view invariance. Distillation reduces cross-view prediction spread by roughly 50% on observable morphological traits (cutting insertion height error by 59% and branching angle error by 60%),

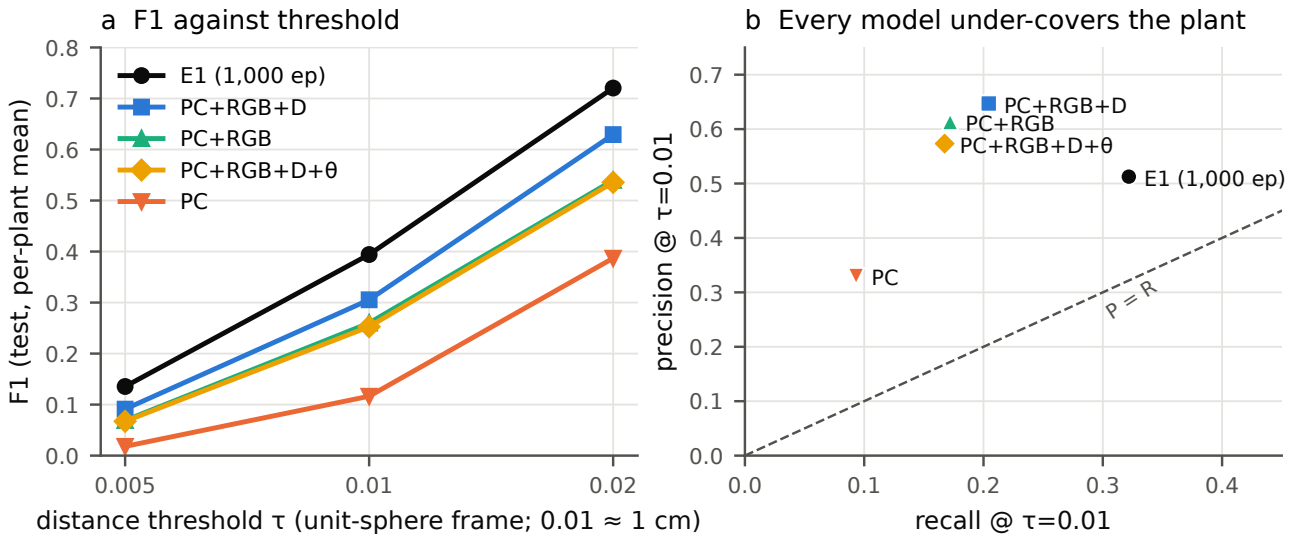


Figure 16. **Point-cloud completion precision, recall, and F_1 score (sorghum test set). (a)** F_1 score across distance thresholds τ. **(b)** Precision versus recall at $\tau = 0.01$, showing that precision consistently exceeds recall on thin plant foliage.

Table 13. **Extended dataset and model capacity scaling (validation set).** Evaluated at a fixed budget of 197,400 gradient steps. Scaling to ViT-Large hurts both species under fixed step compute, while ViT-Small achieves the strongest sorghum roll and leaf-length scores.

	Sorghum		Maize	
Dataset Scale / Model Capacity	Chamfer ($\times 10^{-3}$)	Height R^2	Chamfer ($\times 10^{-3}$)	Height R^2
1,000 Plants	3.13	0.952	3.54	0.393
3,000 Plants	2.86	0.932	1.46	0.524
10,000 Plants	2.48	0.962	0.87	0.427
10,500 Plants (Full Dataset)	2.53	0.962	0.99	0.635
ViT-Small (25 M parameters)	2.86	0.964	1.22	0.563
ViT-Base (114 M parameters)	2.53	0.962	0.99	0.635
ViT-Large (332 M parameters)	2.68	0.948	1.06	0.580

whereas unobservable parameters (such as early-release leaf waviness fields with near-zero skill) become more view-sensitive as the decoder fits image noise, motivating their exclusion from the final parameterization.

G.8. Frozen-Encoder Baselines and Pose Cues

Paired plant-level comparisons confirm that three-stream CROPCODE outperforms EmbodiedMAE-B on 8 of 9 sorghum traits (height $+0.005$, roll $+0.04$, leaf length $+0.08$, leaf width $+0.09$), while four-stream CROPCODE outperforms EmbodiedMAE-B on 10 of 11 maize traits. Under realistic gravity-aligned camera poses (which remove canonical azimuth leakage from synthetic maize renders), Point-MAE height R^2 drops from 0.720 to 0.588 while pose-free CROPCODE wins 5 of 9 decided traits.

G.9. Masked Reconstruction Against Published Encoders

Table 14 compares masked input reconstruction against published decoder-equipped encoders on view 00 across all 2,250 validation plants per species. For image reconstruction, 75% of RGB and depth patches are masked and the point cloud is omitted; three-stream CROPCODE outperforms MultiMAE on hidden RGB PSNR in both species (28.0 vs. 24.7 dB in sorghum; 19.7 vs. 14.4 dB in maize), while MultiMAE achieves lower depth RMSE in sorghum (280 vs. 423 mm). For point-cloud completion (60% of point groups masked, no image inputs), Point-MAE achieves lower Chamfer and higher F_1 largely because it copies the unmasked 40% input points directly into its output and receives ground-truth group centers, whereas CROPCODE generates the entire cloud from scratch without group centers.

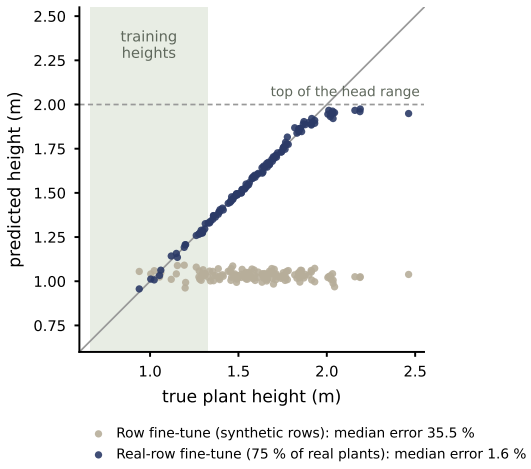


Figure 17. **Predicted versus true plant height on 130 held-out real maize plants** (median across each plant's 36 camera stops using all three input streams). Trained only on synthetic rows, predictions saturate near 1.03 m; after fine-tuning on 75% of the real plants, predictions track true height accurately up to the 2.0 m recipe ceiling. The shaded region indicates the synthetic training height range.

Table 14. **Masked reconstruction against published encoders (validation set, 2,250 plants per species). Image block:** 75% of RGB and depth patches masked and point cloud hidden; PSNR (dB) and depth RMSE/MAE (mm) are evaluated on masked plant pixels (the final row fills masked pixels with mean visible plant depth across 300 plants). **Cloud block:** 60% of point-cloud groups masked with no image input; Chamfer ($\times 10^{-3}$) and F_1 are computed on 2,048 points. Point-MAE retains the visible 40% points and receives hidden group centers, whereas CROPCODE receives neither.

	Sorghum			Maize		
Image block	PSNR ↑	Depth RMSE ↓	Depth MAE ↓	PSNR ↑	Depth RMSE ↓	Depth MAE ↓
MultiMAE-B (RGB+D)	24.7	**280**	**184**	14.4	**215**	150
CROPCODE PC+RGB+D	**28.0**	423	245	**19.7**	216	**145**
CROPCODE four streams	26.9	483	299	18.3	263	183
CROPCODE headline	23.4	400	229	18.1	280	200
Mean visible plant depth	–	336	258	–	236	174
Cloud block	Chamfer ↓	F1@0.01 ↑	F1@0.02 ↑	Chamfer ↓	F1@0.01 ↑	F1@0.02 ↑
Point-MAE (PC)	**1.14**	**0.257**	**0.645**	**0.67**	**0.516**	**0.826**
CROPCODE PC only	9.58	0.044	0.225	4.95	0.061	0.230
CROPCODE PC+RGB+D	7.73	0.069	0.250	4.13	0.080	0.276
CROPCODE four streams	8.50	0.067	0.251	4.57	0.071	0.267
CROPCODE headline	5.93	0.134	0.351	2.94	0.161	0.431

G.10. Latent Space Structure Analysis (E9)

Frozen `[CLS]` embeddings form smooth, continuous manifolds rather than discrete clusters (silhouette score 0.15–0.24, adjusted Rand index ≈ 0.0 against morphological deciles, t-SNE trustworthiness 0.95–0.99). Effective participation ratio is compact—4.4–6.3 of 768 dimensions in sorghum and 6.7–11.3 in maize—with 14–15 principal components explaining 90% of variance in sorghum and 18–19 in maize.

Table 15. **Effect of training occlusion realism on occluded maize reconstruction** (evaluated on 2,250 rendered test rows at a 10 mm threshold).

Training occlusion	F1 ↑	Precision ↑	Recall ↑
Plant alone	0.416	0.460	0.391
Bézier leaves	0.552	0.639	0.507
Synthetic neighbours	0.717	0.764	0.691
Rendered row	**0.789**	**0.804**	**0.782**

Table 16. **Effect of post-training mask ratio on occluded sorghum row reconstruction** (2,250 test plants, evaluated at 10 mm with matching train/test mask ratios).

Mask ratio	F1 ↑	Precision ↑	Recall ↑
0.0	**0.576**	**0.606**	**0.551**
0.2	0.555	0.590	0.526
0.4	0.527	0.568	0.494
0.6	0.475	0.529	0.436
0.8	0.379	0.464	0.325

G.11. Source of Occlusion

Because physically simulated row rendering is computationally expensive, we test whether simpler synthetic occlusions can substitute during fine-tuning (Fig. 22). We compare four training regimes on maize (200 epochs, batch size 32, learning rate 1.5×10^{-4}, 80% masking, QAL $+0.5\times$Sinkhorn loss) evaluated on 2,250 rendered test rows at a 10 mm threshold (Tab. 15 and Fig. 23). Training on isolated plants alone yields an F_1 of 0.416; adding 2D Bézier leaf occluders lifts F_1 to 0.552; placing unbent 3D neighbor point clouds beside the target reaches 0.717; and training on physically contact-solved rendered rows achieves the highest performance at 0.789 (0.804 precision, 0.782 recall).

G.12. Mask Ratio During Post-Training

After masked pretraining and initial row training at mask ratio $m = 0.8$, we fine-tune sorghum row models for 200 epochs at mask ratios $m \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$ and evaluate each at its matching ratio (Tab. 16 and Fig. 24). Because representation learning has already occurred during masked pretraining, lowering the mask ratio during row fine-tuning steadily improves reconstruction by preserving visible evidence of the occluded target plant, lifting $F_1@10\,\text{mm}$ from 0.379 at $m = 0.8$ to 0.576 at $m = 0.0$.

G.13. Point-Cloud Loss

To prevent point-cloud decoders from collapsing points onto central stalks, we combine our quality-aware loss (QAL) with a debiased Sinkhorn divergence. Given predicted cloud $\mathcal{P}$, target cloud $\mathcal{T}$, and nearest-neighbor Euclidean distance $d(x, \mathcal{S}) = \min_{s\in\mathcal{S}} \|x - s\|_2$, QAL weights errors smoothly

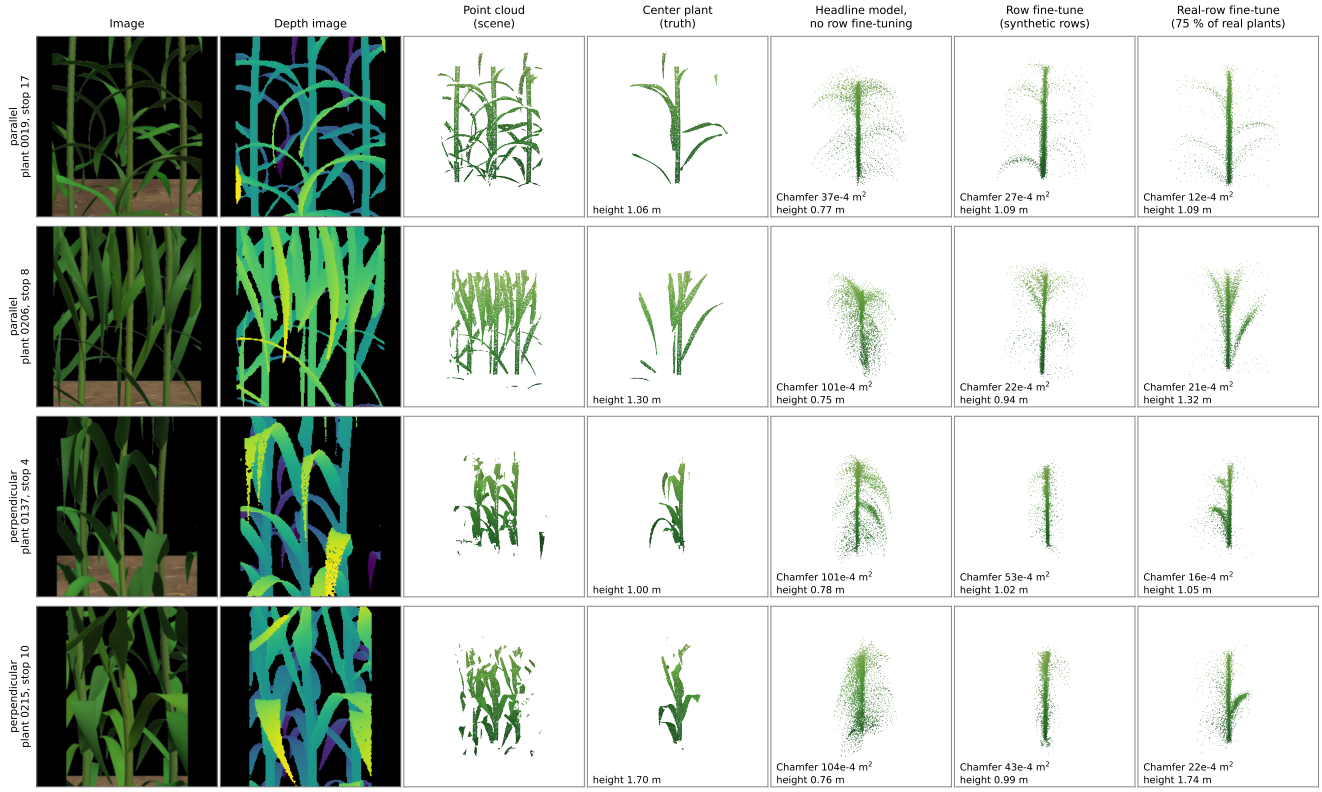


Figure 18. **Qualitative results on held-out real maize rows (E10).** Each row shows a median-error camera stop for parallel and perpendicular row layouts. From left to right: input RGB, depth map, and scene point cloud; visible ground-truth center plant; and predictions from the undistilled headline model, the synthetic row fine-tuned model, and the real-row fine-tuned model, annotated with Chamfer distance (10^{-4} m^2) and predicted height.

around tolerance τ:

$$\mathcal{L}_{\mathrm{QAL}} = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} w\big(d(p,\mathcal{T})\big)\, d(p,\mathcal{T}) + \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} w\big(d(t,\mathcal{P})\big)\, d(t,\mathcal{P}), \tag{8}$$

$$w(d) = \sigma\big(\alpha\,(d-\tau)\big), \tag{9}$$

with $\alpha = 100$ and $\tau = 0.01$ (≈ 10 mm at median plant scale). Trained from scratch for 200 epochs on isolated sorghum plants (Tab. 17), QAL alone achieves high precision (0.561) but low recall (0.144), whereas Sinkhorn divergence alone (S_ε on 1,024 points, blur 0.02) achieves higher recall (0.206) but poor precision (0.142). Combining $\mathcal{L}_{\mathrm{QAL}} + 0.5\,S_\varepsilon$ yields the highest F_1 at both 5 mm (0.083) and 10 mm (0.259), outperforming QAL alone on 82% of test plants. Hyperparameter screening (Tab. 18) confirms that $\tau \in \{5, 10\}$ mm and sharp weighting ($\alpha \in \{100, 300\}$) perform best.

Table 17. **Point-cloud loss ablation on isolated sorghum plants** (trained from scratch, 2,250 test plants).

Loss	F1@5	F1@10	Prec.@10	Rec.@10
QAL	0.057	0.224	**0.561**	0.144
Sinkhorn	0.045	0.168	0.142	**0.206**
QAL + Sinkhorn	**0.083**	**0.259**	0.422	0.189

Table 18. **QAL tolerance τ and sharpness α screen** (isolated sorghum, 3,000 training plants, epoch 300, 225 validation views; @10 denotes 0.01 unit-sphere radius ≈ 10 mm). The $\tau = 10$ mm, $\alpha = 100$ entry is evaluated at epoch 300 of a 300k-step cosine schedule.

τ	α	F1@10	Prec.@10	Rec.@10
5 mm	100	0.155	0.398	0.107
10 mm	100	0.154	0.398	0.107
20 mm	100	0.139	0.384	0.097
10 mm	30	0.146	0.403	0.101
10 mm	300	**0.164**	**0.406**	**0.115**

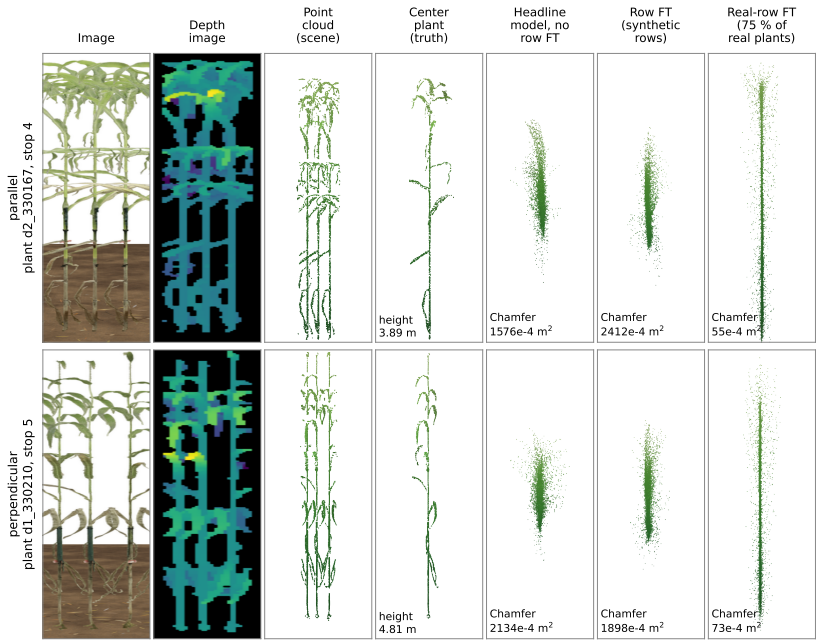


Figure 19. **Qualitative results on held-out real sorghum rows (E10).** Columns follow Fig. 18, with the camera photograph framed vertically to capture the full 2.2–5.3 m plant height (far taller than the 1.74 m synthetic training ceiling). Without real training examples, models predict short, compact plants; fine-tuning on 19 real plants recovers the full stalk height, though fine leaf detail remains limited.

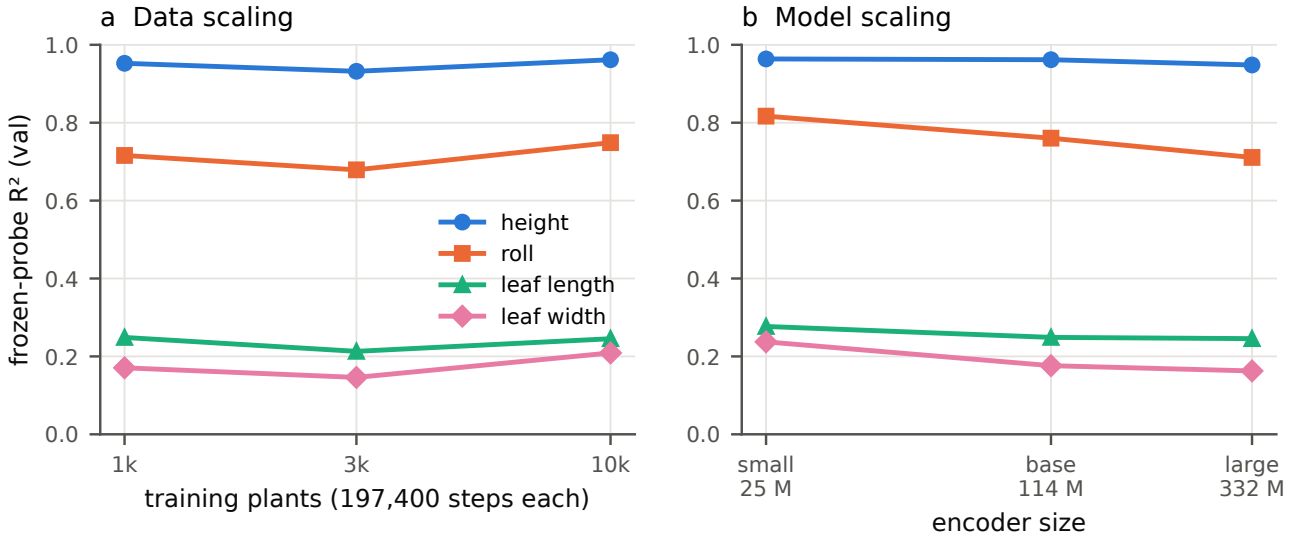


Figure 20. **Dataset volume and model capacity scaling under equal gradient-step budgets (sorghum).** Validation linear-probe R^2 plotted against **(a)** training dataset size and **(b)** encoder parameter count.

G.14. Mixed Masking

We also tested *mixed* masking during occluded training (Fig. 25), where each occluded training scene is masked either by preferentially hiding occluder tokens (with probability 0.5) or uniformly at random (with probability 0.5), evaluated under standard random masking at test time. As shown in Tab. 19, mixed masking improves F_1@10 mm on simplified synthetic neighbor scenes ($+0.048$ in maize, $+0.013$ in sorghum) but offers no advantage over uniform random masking on physically simulated crop rows (-0.003 in maize, -0.004 in sorghum).

Table 19. **Mixed masking versus uniform random masking on occluded test rows** (2,250 test plants). ΔF1@10 denotes mixed minus random.

			Random		Mixed		
Training occlusion	Tested on	Species	F1@5	F1@10	F1@5	F1@10	ΔF1@10
Synthetic scenes	synthetic scenes	maize	0.160	0.439	0.190	**0.487**	+0.048
Synthetic scenes	synthetic scenes	sorghum	0.112	0.310	0.119	**0.323**	+0.013
Rendered rows	rendered rows	maize	0.525	**0.775**	0.522	0.771	−0.003
Rendered rows	rendered rows	sorghum	0.133	**0.377**	0.133	0.373	−0.004

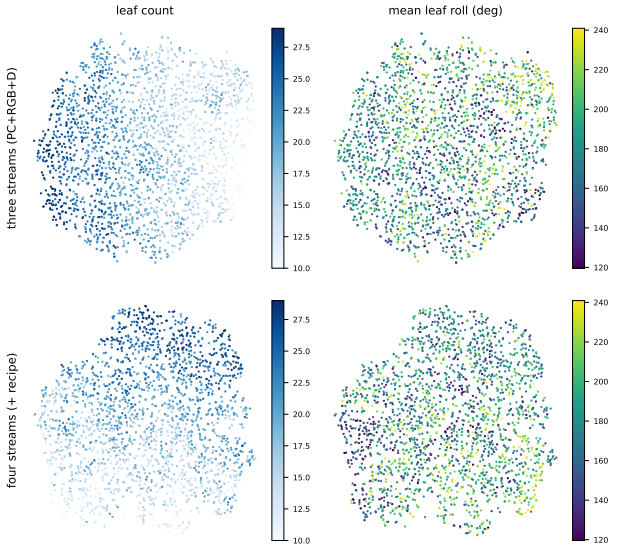


Figure 21. **Latent space structure (sorghum validation set).** t-SNE embeddings of frozen `[CLS]` tokens for the 3-stream (PC + RGB + D) and 4-stream (+θ) models, colored by leaf count and mean leaf roll.

H. Real-Row Fine-Tuning Gallery (E10)

Figures 26 and 27 display 20 additional held-out real plants from the E10 real-row experiment (Sec. 4.8). For maize (Fig. 26), examples are sampled at the 10th, 30th, 50th, 70th, 90th, and 98th error percentiles across both parallel and perpendicular row layouts. Fine-tuning on 75% of the real plants reduces Chamfer error across every percentile (from 20–195 down to 6–52 $\times 10^{-4}$ m^2) and brings predicted heights from a saturated $\sim$ 1.0 m baseline to within a few centimeters of ground truth (e.g., 1.52 vs. 1.52 m and 1.88 vs. 1.85 m). For sorghum (Fig. 27), all eight held-out test plants (2.8–4.9 m tall) improve from 661–2,781 down to 52–122 $\times 10^{-4}$ m^2, recovering full stalk height despite the extreme height extrapolation beyond the 1.74 m synthetic training range.

I. Row Occlusion Gallery (Regime III)

Figures 28 and 29 show ten additional held-out synthetic test plants per species sampled uniformly at the 5th, 15th, . . . , 95th percentiles of row-trained Chamfer error across broadside and end-on views.

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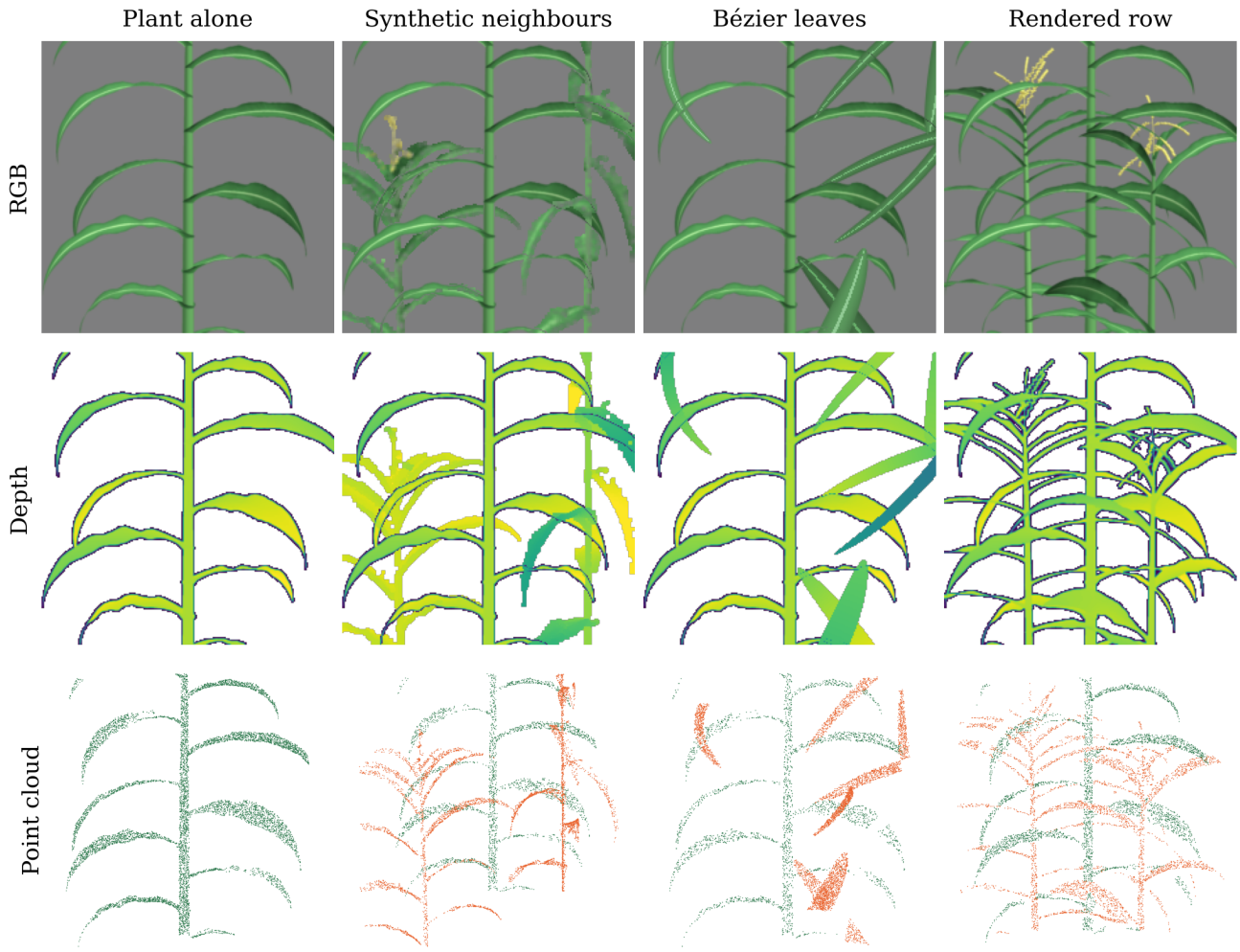


Figure 22. **Comparison of four training occlusion sources on maize.** Columns: isolated target plant, synthetic unbent 3D neighbors, synthetic Bézier leaf occluders, and physically contact-solved rendered rows. Rows: RGB image, metric depth map, and 3D point cloud (target points in green, occluders in orange).

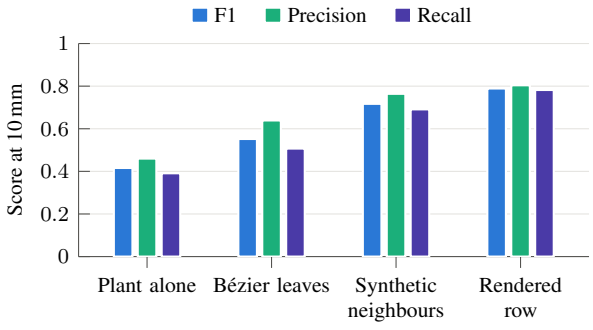


Figure 23. **Occluded maize reconstruction across training occlusion sources** (data from Tab. 15). F_1, precision, and recall increase monotonically as training occlusion approaches physical row realism.

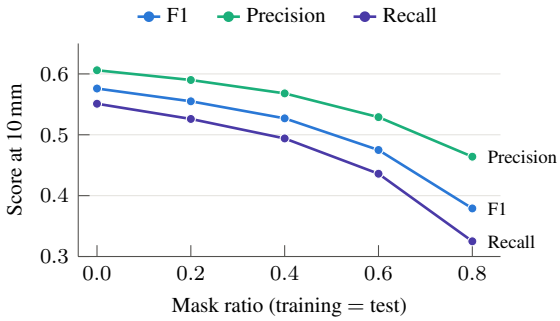


Figure 24. **Occluded row reconstruction versus post-training mask ratio** (data from Tab. 16). Keeping all visible tokens unmasked during row fine-tuning maximizes F_1, precision, and recall.

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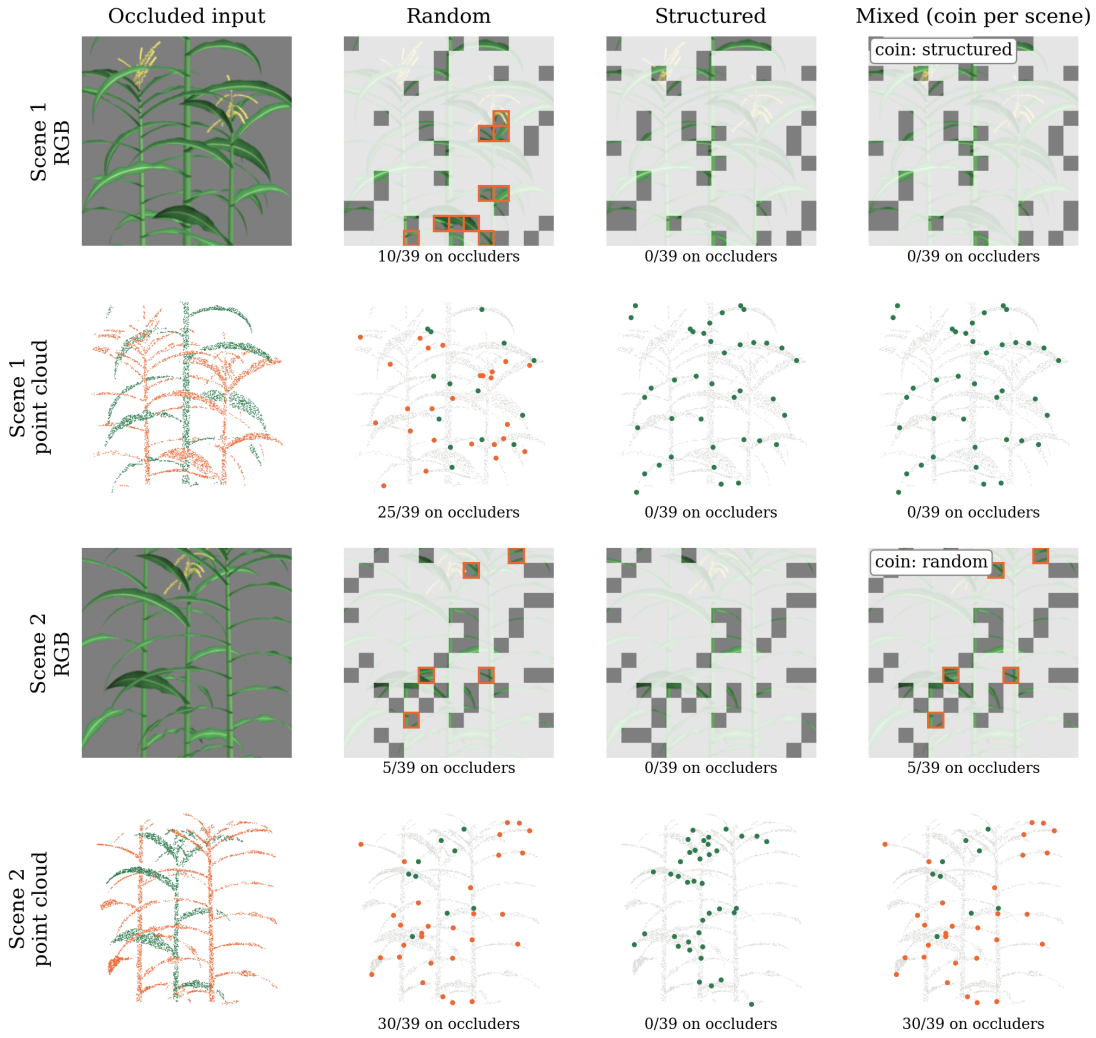


Figure 25. **Comparison of random, structured, and mixed masking policies on two occluded maize row scenes.** Structured masking prioritizes hiding tokens dominated by neighboring plants, whereas mixed masking flips a coin per scene between structured and random masking. Visible tokens are shown at full contrast, with occluder-overlapping tokens outlined in orange.

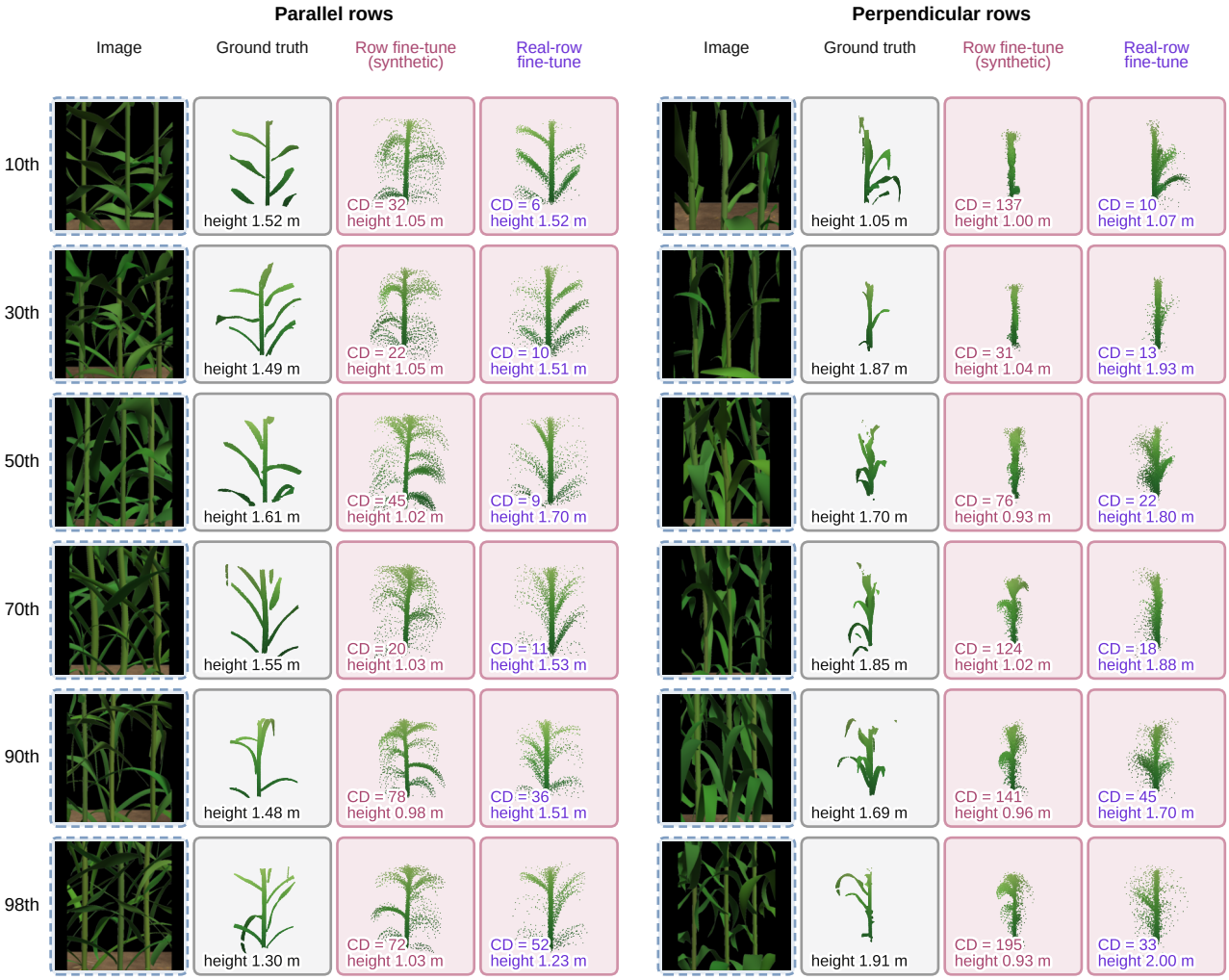


Figure 26. **Held-out real maize rows across error percentiles (E10).** Parallel (left) and perpendicular (right) row layouts sampled from the 10th to the 98th percentile of real-row fine-tuned Chamfer error. Columns show the input RGB image, ground-truth center plant with true height, synthetic row fine-tuned prediction, and real-row fine-tuned prediction with metric Chamfer distance (10^{-4} m^2) and predicted height.

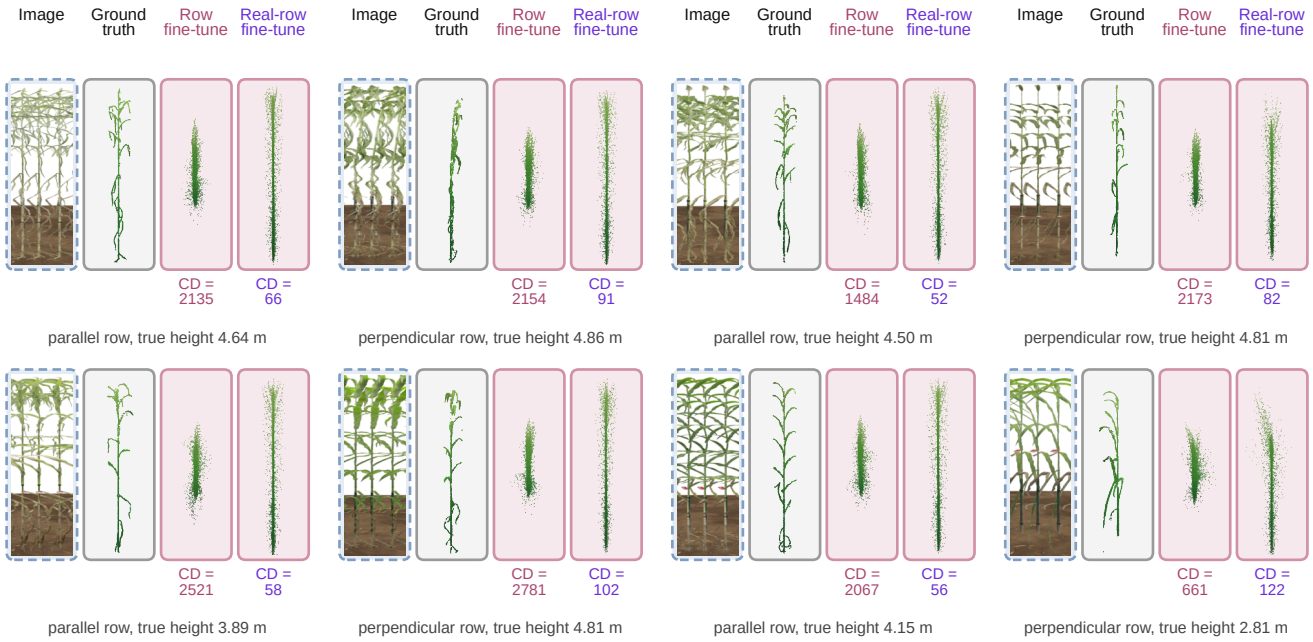


Figure 27. **All eight held-out real sorghum plants in crop rows (E10).** Center camera stops with alternating row layouts; columns follow Fig. 26, with true plant height noted beneath the ground-truth cloud.

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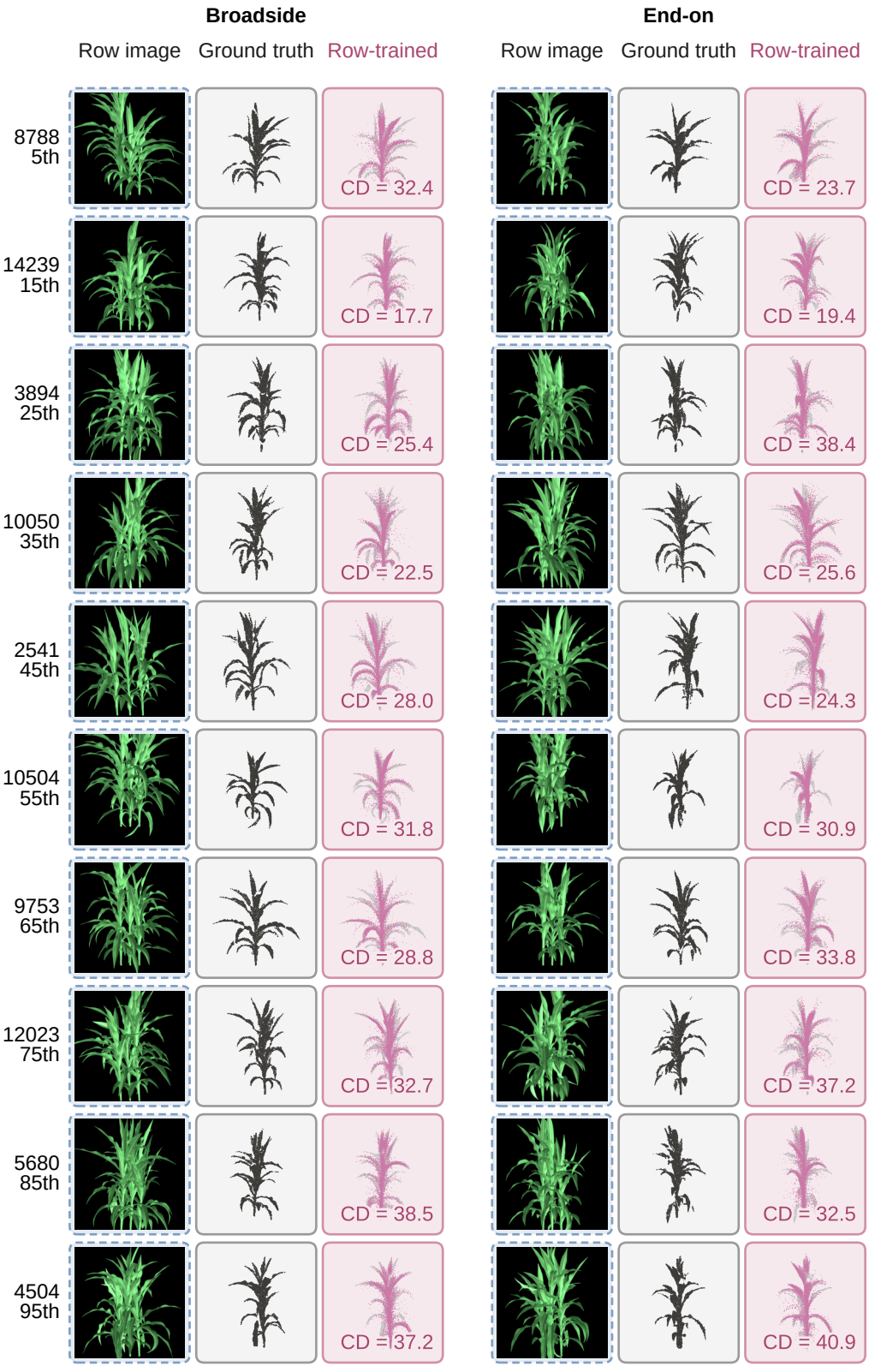


Figure 28. **Sorghum test rows sampled evenly across the error distribution (5th to 95th percentiles).** Broadside (left) and end-on (right) views showing the input row photograph, ground-truth center plant, and row-trained prediction (pink) overlaid on ground truth (grey), with metric Chamfer distance (10^{-4} m^2).

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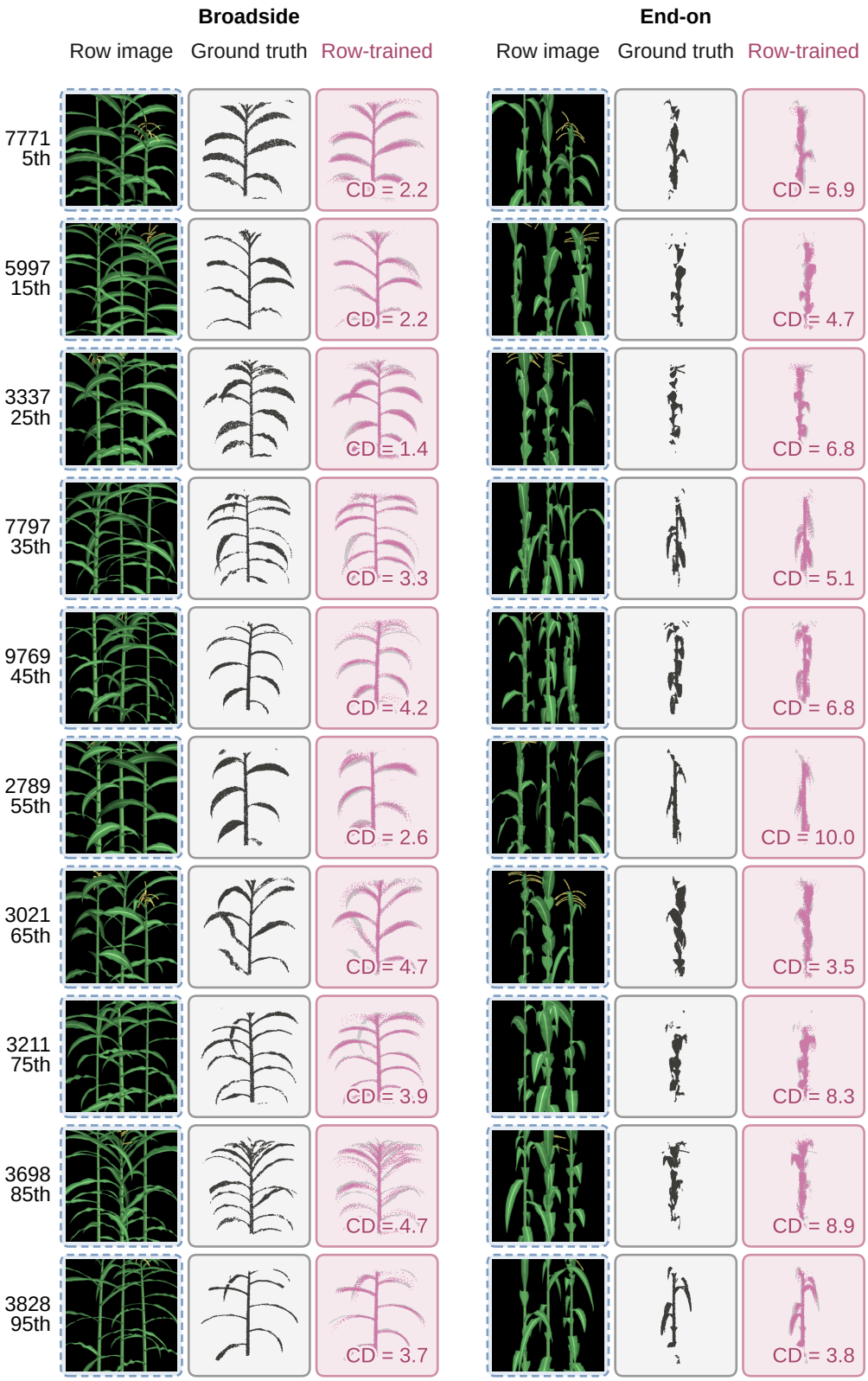


Figure 29. **Maize test rows sampled evenly across the error distribution (5th to 95th percentiles).** Columns and layout follow Fig. 28.

J. Dataset Generation Details

FIELDSCAN comprises four synthetic subsets: isolated single-plant captures and physically contact-solved simulated rover rows for both maize and sorghum. All renders are generated at 1024×1024 resolution with a 40° vertical field of view ($f_x = f_y = 512/\tan 20^\circ = 1406.7$ px, principal point at image center) in OpenGL camera coordinates ($+y$ up, camera looking down $-z$).

J.1. Procedural Crop Generation

Maize procedural generator. Each of the 15,000 maize plants is synthesized from random seed i using a procedural maize model [48]. Base traits are drawn from Gaussian distributions: plant height 0.832 ± 0.104 m, leaf count 12.35 ± 2.57, base leaf length 0.448 ± 0.057 m, base leaf width 0.066 ± 0.010 m, and base insertion angle $43.98^\circ \pm 10.13^\circ$. Traits are coupled through four latent factors (vigor, slenderness, posture, and texture; correlation strength 0.735), canopy position curves (bottom, middle, top), and per-leaf spatial jitter (scale 0.687), producing plant heights from 0.62 to 1.44 m (median 1.01 m). Each record stores global height, leaf count, and per-leaf insertion height, length, width, azimuth, insertion angle, and droopiness, with midrib curvature integrated via 10-node Gauss–Legendre quadrature along B-spline centerlines.

Sorghum procedural generator. Each of the 15,000 sorghum plants is generated from seed i using Evo-Engine [33]. Heading angle is sampled uniformly in $[0^\circ, 360^\circ)$, stem tilt $0^\circ \pm 1.5^\circ$, leaf count 20 ± 2 (clamped to $[2, 128]$), and internode spacing 0.04 ± 0.03 m, yielding stem lengths internode $\times\, n_{\text{leaves}}/(1 - s_0)$ between 0.19 and 1.74 m across 4–35 leaves (mean 19.5). Continuous curves over normalized stem height $t \in [0, 1]$ govern leaf insertion height, length, width, alternating roll ($\text{roll}_k = (k \bmod 2) \cdot 180^\circ + \text{roll}(t)$), branching angle, and curvature.

J.2. Dataset Partitioning

Both species are split 70% / 15% / 15% into 10,500 training, 2,250 validation, and 2,250 test plants, enriching validation and test splits toward extreme phenotypes:

- **Maize:** 19 morphological features are standardized via median absolute deviation ($z = (x - \text{median})/(1.4826 \cdot \text{MAD})$) and scored by Mahalanobis distance $\sqrt{z^\top (\Sigma + 10^{-6} I)^+ z}$. The most extreme plants are assigned alternately to test and validation splits (train scores 1.56–4.60; validation 4.60–23.3; test 4.60–32.2).
- **Sorghum:** Plants are ranked to percentile $r \in [0, 1]$ by Mahalanobis distance across six standardized traits (leaf count, stem length, mean and maximum leaf length, and mean and standard deviation of leaf roll). A 30% evaluation pool is sampled without replacement with probability proportional to $(r + 0.02)^2$ (seed 42) and split equally into validation and test sets (median rank 0.75 vs. 0.37 in train), deterministically reproduced by `data_split/make_split.py`.

Figure 30 illustrates the resulting split distributions, inter-trait correlations, and row occlusion statistics.

J.3. Isolated-Plant Sensor Rendering

Plants are rendered off-screen in `pyrender` aimed at bounding-box center $\mathbf{c}$ with 0.15 ambient light and three directional lights from $(1, 1, 1)$, $(-1, 0.5, -0.5)$, and $(0, 1, 0)$ (intensities 4.0, 2.0, and 1.5) against a neutral grey background (0.5). Metric depth $z \in [n, f]$ is packed losslessly into 32-bit RGBA PNG channels:

$$v = \left\lfloor \text{clip}\left(\frac{z - n}{f - n}, 0, 1\right)(2^{32} - 1) \right\rfloor, \qquad (10)$$

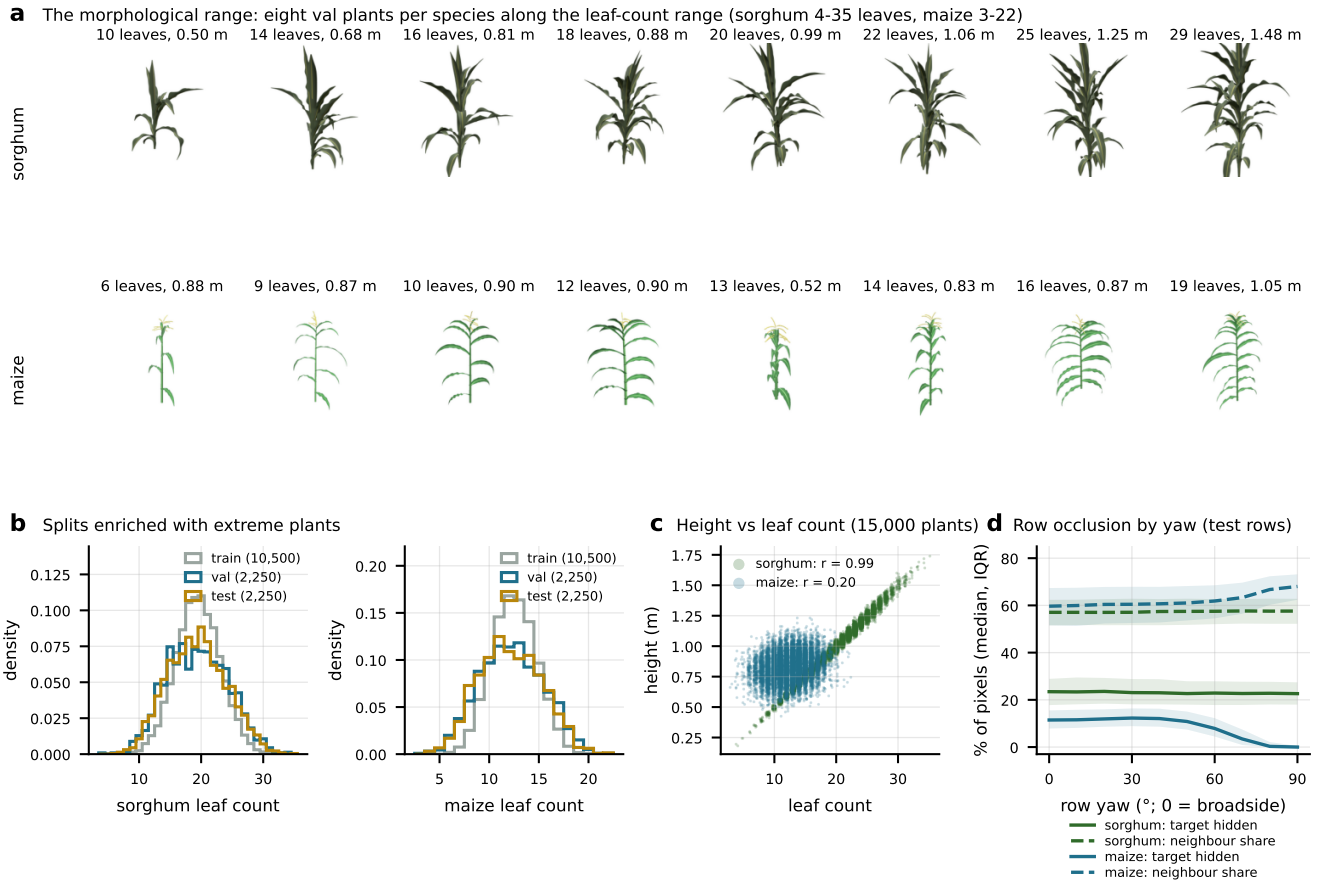


Figure 30. **Statistical overview of FieldScan. (a)** Eight validation plants per species spanning the leaf-count range. **(b)** Leaf-count distributions across train, validation, and test splits, illustrating tail enrichment in validation and test. **(c)** Plant height versus leaf count across all 15,000 plants per species ($r = 0.99$ in sorghum vs. $r = 0.20$ in maize). **(d)** Row occlusion across yaw angles on test rows, showing the fraction of target pixels occluded by neighbors (solid) and the fraction of total plant pixels belonging to neighbors (dashed), plotted as median and interquartile range.

and valid pixels (u, v) back-project to 3D camera coordinates via:

$$\mathbf{p}_{\text{cam}} = \left(\frac{u - c_x}{f_x} z,\ -\frac{v - c_y}{f_y} z,\ -z \right). \tag{11}$$

Maize captures. Ten camera viewpoints are sampled uniformly on the unit sphere at bounding-box-fitted distance:

$$\text{dist} = \frac{1.05\, r}{\sin 20^\circ}, \qquad \begin{aligned} n &= \max(0.001,\ \text{dist} - 1.5\, r), \\ f &= \text{dist} + 1.5\, r, \end{aligned} \tag{12}$$

where r is half the bounding-box diagonal, with a fourth camera-mounted light (intensity 1.5). Each view outputs RGB, depth, camera pose, procedural parameters, and an 8,192-point cloud back-projected from the depth map. Per-leaf curvature is integrated along the rebuilt midrib B-spline over arc length $s \in [0.22, 1]$:

$$\theta = \int_{0.22}^{1} |\kappa(s)|\, ds, \qquad \bar{\kappa} = \theta / S. \tag{13}$$

Sorghum captures. Ten viewpoints lie on a fixed Fibonacci sphere rig at 2.5 m distance ($y_i = 1 - 2(i + 0.5)/10$, $\phi_i = i(\sqrt{5} - 1)\pi + \epsilon_i$, spanning elevations from -64° to $+64^\circ$), outputting RGB, depth, camera pose, procedural parameters, and complete occlusion-free 8,192-point clouds in the camera frame.

J.4. Simulated Crop Row Rendering

Row layout and articulated contact solver. Each row places target plant L_i between right neighbor L_{i+1} and left neighbor L_{i+2} at stem spacing $d = 0.15$ m along the x-axis across ten row orientations $\theta \in \{0^\circ, 10^\circ, \ldots, 90^\circ\}$ with neighbor yaw jitter $\delta \sim U(-15^\circ, 15^\circ)$ (sorghum neighbors

are also aligned so their principal SVD stem axes run parallel to the target stem). To prevent unrealistic mesh interpenetration, our articulated contact solver models leaf blades as chains of 8 cm jointed segments while keeping stems, sheaths (within 2.5 cm of the stem), and maize tassels rigid.

Contacts are detected via a two-stage MEETS check: a 1 cm dilated voxel grid screen ($\varepsilon_O = 2.8$ mm, $\varepsilon_M = 6.8$ mm) followed by exact triangle–triangle intersection tests (allowing surfaces to touch without crossing). As detailed in Algorithms 1–3 and Table 20, target leaves bend only to clear rigid neighbor stems/sheaths, neighbors yield to the target and each other in iterative rounds (including stem-axis rolls up to $\pm 80^\circ$ for blocked sorghum neighbor leaves), and target leaves bend round neighbor foliage only if a neighbor leaf remains boxed in. If collisions persist, joint limits loosen across three stages and stem spacing widens in 5 cm steps (refined to 1 cm) up to 1.5 m. Finally, three settling passes ease every bent leaf back toward its grown shape via 12 binary halvings so leaves rest naturally within a fraction of a millimeter of adjacent foliage. Across all 300,000 row orientations (15,000 rows $\times$ 10 yaws per species), the solver converged for every orientation, and its final exact triangle–triangle pass accepted each layout only with no inter-plant crossings (Table 21). An independent intersection-check suite sharing no code with the solver also found no crossings on the rows it was run on; which rows those were was not recorded, and the bent meshes are not stored with the renders, so that check cannot be repeated on the released data.

Table 20. **Articulated contact solver joint bending limits.** Staged angular limits used to resolve inter-plant leaf collisions in simulated crop rows.

Stage	Collar side	Blade side	Up/down	Total turn
Stage 0 (default)	30°	10°	45°	90°
Stage 1 (medium)	60°	10°	60°	120°
Stage 2 (loose)	90°	30°	70°	180°

Table 21. **Crop row contact-solver convergence statistics.** "Bent" denotes orientations where at least one target leaf bent; "Rolled" denotes orientations where at least one neighbor leaf rolled. No orientation failed, and the solver's final exact triangle–triangle pass found no inter-plant crossings.

Split	Orient.	At d=0.15 m	Stage 0 / 1 / 2	Max / p99 d	Bent	Rolled
Maize train	105,000	76.4%	81,679 / 16,138 / 7,183	0.33 / 0.23 m	48.0%	–
Maize val	22,500	78.3%	17,780 / 3,321 / 1,399	0.28 / 0.23 m	46.8%	–
Maize test	22,500	79.0%	17,824 / 3,284 / 1,392	0.30 / 0.23 m	47.1%	–
Sorghum train	105,000	85.4%	42,272 / 25,120 / 37,608	0.39 / 0.21 m	98.3%	99.4%
Sorghum val	22,500	86.0%	9,395 / 5,335 / 7,770	0.63 / 0.21 m	97.8%	98.9%
Sorghum test	22,500	86.3%	9,511 / 5,416 / 7,573	0.46 / 0.21 m	97.3%	99.1%

Rover camera capture and dataset notes. Each row orientation is captured by a horizontal rover camera at $(0, h, \mathrm{dist})$ looking at $(0, h, 0)$, where h is the target bounding-box center height ($\mathrm{dist} = 1.0$ m for maize, 2.5 m for sorghum),

Algorithm 1 BENDCLEAR: Minimum-deviation collision-free leaf shape

Require: leaf ℓ with segments $0, \ldots, n-1$ (collar first); obstacles $\mathcal{O}$; stage limits (U, S_0, S, T)
$p \leftarrow$ grown shape (every joint bend $(\alpha_j, \beta_j) = (0, 0)$)
$j \leftarrow$ first segment of $\ell(p)$ that MEETS $\mathcal{O}$ ▷ none: clear
while j exists **do**
  if too many steps or test budget spent **then return** FAIL
  $q^\star \leftarrow$ none
  for $k = j, j-1, \ldots, \max(0, j-3)$ **do** ▷ joint before contact first
    for all bends (α, β) of joint k, smallest first **do**
      $q \leftarrow p$ with joint k set to (α, β), joints after k straightened
      if $\sum_i \|(\alpha_i, \beta_i)\| > T$ **or** segs. $k..j$ of $\ell(q)$ MEETS $\mathcal{O}$ **then**
        continue
      $q \leftarrow$ least bend between p and q at joint k that clears (3 halvings)
      if $\ell(q)$ enters own stem or crosses own plant more than as grown **then**
        continue
      $q^\star \leftarrow q$; **break** both loops
  if $q^\star$ is none **then return** FAIL
  $p \leftarrow q^\star$; $j \leftarrow$ first segment $\geq k$ of $\ell(p)$ that MEETS $\mathcal{O}$
return p

For a sorghum neighbor leaf, on FAIL the search repeats from the grown shape rolled about the stem by $\pm 20^\circ, \pm 40^\circ, \ldots, \pm 80^\circ$, keeping the first clear roll.

Algorithm 2 SOLVELAYOUT: Iterative contact resolution for a row layout

Require: main plant P_0, neighbors P_1, P_2 at spacing d, angles θ; stage limits
if stems or sheaths of any two plants cross **then return** FAIL
for all leaves ℓ of P_0 that MEETS the stems/sheaths of P_1, P_2 **do**
  $\ell \leftarrow$ BENDCLEAR(ℓ, stems and sheaths of P_1, P_2)
  if FAIL **return** FAIL
for round $= 1, \ldots, 8$ **do** ▷ neighbors give way first
  for $r = 1, 2$ **do**
    for all leaves ℓ of P_r that MEETS the other two plants **do**
      $\ell \leftarrow$ BENDCLEAR(ℓ, other two plants); on FAIL mark ℓ stuck
  if no leaf changed **then break**
for $t = 1, 2$ **while** some neighbor leaf is stuck **do** ▷ target yields only if necessary
  bend clear the P_0 leaves that MEETS a stuck leaf; rerun the neighbor rounds
return OK if no triangle of any plant crosses another plant (exact test), else FAIL

producing both the three-plant row input (RGB, depth, and 24,576-point scene cloud) and the isolated contact-bent target ground truth (8,192-point cloud and procedural parameters). Note two minor edge cases in the dataset: (1) at 2.5 m distance with a 40° vertical field of view (≈ 1.8 m vertical span), the top or base of the tallest sorghum plants can extend slightly outside the image frame; and (2) sorghum training plant 0 (generated with seed 0, where EvoEngine does not reseed its RNG) omits parameters for 2 of its 13 leaves and has a mesh 0.35 m taller than predicted by its stem length,

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Algorithm 3 SOLVEORIENTATION: Spacing search, stage progression, and settling

Require: plants P_0, P_1, P_2; angles θ; $d_0 = 0.15\,\text{m}$, $d_{\max} = 1.5\,\text{m}$

function TRY(d)
  for stage $s = 0, 1, 2$ **do**
    $r \leftarrow$ SOLVELAYOUT$(P_0, P_1, P_2, d, \theta, \text{limits}_s)$
    if $r =$ OK **or** stems/sheaths cross **then return** r
  return FAIL
if TRY$(d_0) =$ OK **then** $d \leftarrow d_0$
else
  $d \leftarrow$ first of $d_0{+}0.05, d_0{+}0.10, \ldots, d_{\max}$ with TRY$(d) =$ OK
  $d_{\text{lo}} \leftarrow$ last failed spacing tried before d
  $d \leftarrow$ first of $d_{\text{lo}}{+}0.01, d_{\text{lo}}{+}0.02, \cdots < d$ with TRY $=$ OK, else keep d
settle every bent leaf of the layout kept at d (resting against neighbors)
render the input (three plants) and ground truth (target plant alone in the same bent shape)

affecting only a single training instance and no validation or test plants.